

Chapter 12

Interrupt Matrix

12.1 Overview

The interrupt matrix embedded in ESP32-P4 independently routes peripheral interrupt sources to the ESP-RISC-V CPU's peripheral interrupts to timely inform HP CPU0 or HP CPU1 to process the coming interrupts.

Peripheral interrupt sources must be routed to HP CPU0 or HP CPU1 through the interrupt matrix for the following reasons:

- Each HP CPU has 32 interrupt inputs, but the ESP32-P4 provides 128 peripheral interrupt sources. To connect all sources to the CPUs, the interrupt matrix is required.
- The interrupt matrix allows a single peripheral interrupt source to be mapped to any HP CPU0 or HP CPU1 interrupt input, depending on application requirements.

Note:

This chapter focuses on how to map peripheral interrupt sources to HP CPU0/HP CPU1 interrupts. For more details about interrupt configuration, interrupt vectors, and interrupt handling operations recommended by the ISA, please refer to Chapter 1 [High-Performance CPU](#).

12.2 Interrupt Terminology in ESP32-P4

The following terms related to interrupts are defined in the context of the *ESP32-P4 Technical Reference Manual* to help readers better understand this document:

12.2.1 Interrupt

An interrupt is an event or condition that temporarily suspends the CPU's current execution to execute an interrupt service routine (ISR). This mechanism enables the CPU to respond immediately to important events.

The *ESP32-P4 Technical Reference Manual* uses the term “interrupt” in a broader sense to refer to both interrupt signal and interrupt source.

12.2.2 Interrupt Signal/Interrupt Source

Both interrupt signal and interrupt source describe the same signal from different points in the system.

From the perspective of peripherals, interrupt signals are generated by the peripheral's internal interrupt sources and are sent to the interrupt matrix.

From the perspective of the interrupt matrix, it receives the interrupt signals sent from peripherals and considers them interrupt sources. The interrupt matrix then outputs CPU peripheral interrupt signals to the CPU.

From the perspective of the CPU, the interrupt signals from the interrupt matrix become interrupt sources and are sent to the CPU core together with the core local interrupt sources.

12.2.3 Interrupt Flow in ESP32-P4

Figure 12.2-1 shows the interrupt flow in ESP32-P4.

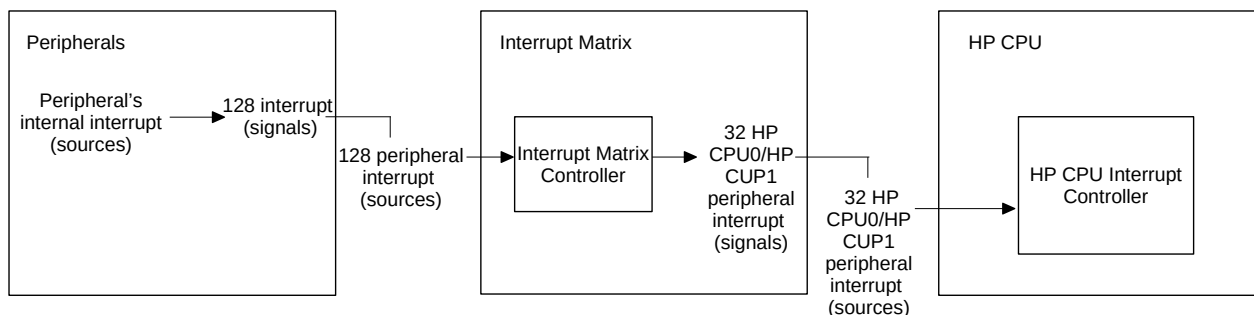


Figure 12.2-1. Interrupt Flow in ESP32-P4

12.3 Features

The interrupt matrix embedded in the ESP32-P4 has the following features:

- 128 peripheral interrupt sources accepted as input
- 32 HP CPU0 peripheral interrupts and 32 HP CPU1 peripheral interrupts generated to HP CPU as output
- Current interrupt status query for peripheral interrupt sources
- Multiple interrupt sources mapped to a single HP CPU0 or HP CPU1 interrupt (i.e., shared interrupts)
- Remapping of HP CPU0 or HP CPU1 User Mode interrupts to Machine Mode interrupts

Figure 12.3-1 shows the structure of the interrupt matrix.

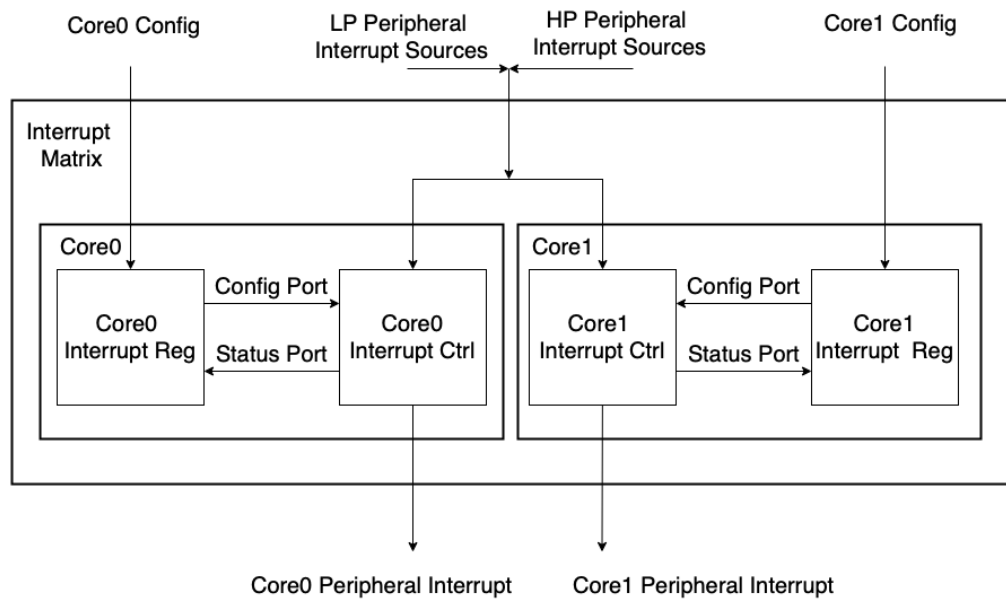


Figure 12.3-1. Interrupt Matrix Structure

All interrupts generated by the peripheral interrupt sources can be handled by HP CPU0 or HP CPU1. Users can configure [HP CPU0 interrupt registers](#) (“Core0 Interrupt Reg” module in Figure 12.3-1) to assign peripheral interrupt sources to HP CPU0, or configure [HP CPU1 interrupt registers](#) (“Core1 Interrupt Reg” module in Figure 12.3-1) to assign peripheral interrupt sources to HP CPU1. Peripheral interrupt sources can be mapped to both HP CPU0 and HP CPU1 at the same time, allowing both to receive the interrupts.

12.4 Functional Description

12.4.1 Peripheral Interrupt Sources

ESP32-P4 has 128 peripheral interrupt sources in total. Table 12.4-1 lists all these sources and their mapping/status registers.

- Column “No.”: Peripheral interrupt source number, which can be 0 ~ 82, 84 ~ 98, 100 ~ 111, 113 ~ 130 (83, 99 and 112 are reserved).
- Column “Chapter”: The chapter in which the interrupt source is described in detail.
- Column “Interrupt Source”: Name of the peripheral interrupt source.
- Column “Interrupt Source Mapping Register”: Registers used for routing the peripheral interrupt sources to HP CPU peripheral interrupts.
- Column “Interrupt Status Register”: Registers used for indicating the interrupt status of peripheral interrupt sources.
 - Column “Interrupt Status Register - Bit”: Bit position in status register, indicating the interrupt status.
 - Column “Interrupt Status Register - Name”: Name of status registers.

The registers in column “Interrupt Source Mapping Register” and column “Interrupt Status Register” correspond to the peripheral interrupt source in column “Interrupt Source”. For example, the interrupt source

mapping register for interrupt source RMT_INTR is `COREx_RMT_INT_MAP_REG`, and its status register is `COREx_INTR_STATUS_REG_1_REG`.

Note that CORE_x in the table can be CORE0 (HP CPU0) or CORE1 (HP CPU1).

Table 12.4-1. CPU Peripheral Interrupt Source Mapping/Status Registers and Peripheral Interrupt Sources

No.	Chapter	Interrupt Source	Interrupt Source Mapping Register	Interrupt Status Register	
				Bit	Name
0	RTC Timer	LP_RTC_INTR	COREx_LP_RTC_INT_MAP_REG	0	COREx_INTR_STATUS_REG_O_REG
1	Watchdog Timers (WDT)	LP_WDT_INTR	COREx_LP_WDT_INT_MAP_REG	1	
2	RTC Timer	LP_TIMER_REG_O_INTR	COREx_LP_TIMER_REG_O_INT_MAP_REG	2	
3	RTC Timer	LP_TIMER_REG_1_INTR	COREx_LP_TIMER_REG_1_INT_MAP_REG	3	
4	LP Mailbox	MB_LP_INTR	COREx_MB_LP_INT_MAP_REG	4	
5	LP Mailbox	MB_HP_INTR	COREx_MB_LP_INT_MAP_REG	5	
6	Low-Power Management	PMU_REG_O_INTR	COREx_PMU_REG_O_INT_MAP_REG	6	
7	Low-Power Management	PMU_REG_1_INTR	COREx_PMU_REG_1_INT_MAP_REG	7	
8	Brown-out Detector	LP_ANAPERI_INTR	COREx_PMU_REG_O_INT_MAP_REG	8	
9	ADC Controller (ADC)	LP_ADC_INTR	COREx_PMU_REG_O_INT_MAP_REG	9	
10	GPIO Matrix and IO MUX	LP_GPIO_INTR	COREx_LP_GPIO_INT_MAP_REG	10	
11	I2C Controller (I2C)	LP_I2C_INTR	COREx_LP_I2C_INT_MAP_REG	11	
12	I2S Controller (I2S)	LP_I2S_INTR	COREx_LP_I2S_INT_MAP_REG	12	
13	SPI Controller (SPI)	LP_SPI_INTR	COREx_LP_SPI_INT_MAP_REG	13	
14	Touch Sensor (TOUCH)	LP_TOUCH_INTR	COREx_LP_TOUCH_INT_MAP_REG	14	
15	Temperature Sensor (TSENS)	LP_TSENS_INTR	COREx_LP_TSENS_INT_MAP_REG	15	
16	UART Controller (UART)	LP_UART_INTR	COREx_LP_UART_INT_MAP_REG	16	
17	eFuse Controller (EFUSE)	LP_EFUSE_INTR	COREx_LP_EFUSE_INT_MAP_REG	17	
18	Low-Power CPU	LP_SW_INTR	COREx_LP_SW_INT_MAP_REG	18	
19	System Registers (SYSREG)	LP_SYSREG_INTR	COREx_LP_SYSREG_INT_MAP_REG	19	
20	Key Manager	LP_HUK_INTR	COREx_LP_HUK_INT_MAP_REG	20	
21	System Registers (SYSREG)	SYS_ICM_INTR	COREx_SYS_ICM_INT_MAP_REG	21	
22	USB Serial/JTAG Controller (USB_SERIAL_JTAG)	USB_DEVICE_INTR	COREx_USB_DEVICE_INT_MAP_REG	22	
23	SD/MMC Host Controller (SDHOST)	SDIO_HOST_INTR	COREx_SDIO_HOST_INT_MAP_REG	23	
24	GDMA Controller (GDMA-AHB, GDMA-AXI)	GDMA_INTR	COREx_GDMA_INT_MAP_REG	24	
25	SPI Controller (SPI)	SPI2_INTR	COREx_SPI2_INT_MAP_REG	25	
26	SPI Controller (SPI)	SPI3_INTR	COREx_SPI3_INT_MAP_REG	26	
27	I2S Controller (I2S)	I2SO_INTR	COREx_I2SO_INT_MAP_REG	27	
28	I2S Controller (I2S)	I2S1_INTR	COREx_I2S1_INT_MAP_REG	28	
29	I2S Controller (I2S)	I2S2_INTR_O	COREx_I2S2_INT_MAP_REG	29	
30	UART Controller (UART)	UHCIO_INTR	COREx_UHCIO_INT_MAP_REG	30	
31	UART Controller (UART)	UARTO_INTR	COREx_UARTO_INT_MAP_REG	31	

No.	Chapter	Interrupt Source	Interrupt Source Mapping Register	Interrupt Status Register	
				Bit	Name
32	UART Controller (UART)	UART1_INTR	COREx_UART1_INT_MAP_REG	0	COREx_INTR_STATUS_REG_1_REG
33	UART Controller (UART)	UART2_INTR	COREx_UART2_INT_MAP_REG	1	
34	UART Controller (UART)	UART3_INTR	COREx_UART3_INT_MAP_REG	2	
35	UART Controller (UART)	UART4_INTR	COREx_UART4_INT_MAP_REG	3	
36	LCD and Camera Controller (LCD_CAM)	LCD_CAM_INTR	COREx_LCD_CAM_INT_MAP_REG	4	
37	ADC Controller (ADC)	ADC_INTR	COREx_ADC_INT_MAP_REG	5	
38	Motor Control PWM (MCPWM)	PWM0_INTR	COREx_PWM0_INT_MAP_REG	6	
39	Motor Control PWM (MCPWM)	PWM1_INTR	COREx_PWM1_INT_MAP_REG	7	
40	Two-Wire Automotive Interface (TWAI)	TWAI0_INTR	COREx_TWAI0_INT_MAP_REG	8	
41	Two-Wire Automotive Interface (TWAI)	TWAI1_INTR	COREx_TWAI1_INT_MAP_REG	9	
42	Two-Wire Automotive Interface (TWAI)	TWAI2_INTR	COREx_TWAI2_INT_MAP_REG	10	
43	Remote Control Peripheral (RMT)	RMT_INTR	COREx_RMT_INT_MAP_REG	11	
44	I2C Controller (I2C)	I2C0_INTR	COREx_I2C0_INT_MAP_REG	12	
45	I2C Controller (I2C)	I2C1_INTR	COREx_I2C1_INT_MAP_REG	13	
46	Timer Group (TIMG)	TIMERGRPO_TO_INTR	COREx_TIMERTGRPO_TO_INT_MAP_REG	14	
47	Timer Group (TIMG)	TIMERGRPO_T1_INTR	COREx_TIMERGRPO_T1_INT_MAP_REG	15	
48	Timer Group (TIMG)	TIMERGRPO_WDT_INTR	COREx_TIMERGRPO_WDT_INT_MAP_REG	16	
49	Timer Group (TIMG)	TIMERGRP1_TO_INTR	COREx_TIMRTGRP1_TO_INT_MAP_REG	17	
50	Timer Group (TIMG)	TIMERGRP1_T1_INTR	COREx_TIMERGRP1_T1_INT_MAP_REG	18	
51	Timer Group (TIMG)	TIMERGRP1_WDT_INTR	COREx_TIMERGRP1_WDT_INT_MAP_REG	19	
52	LED PWM Controller (LEDC)	LEDC_INTR	COREx_LEDC_INT_MAP_REG	20	
53	System Timer	SYSTIMER_TARGET0_INTR	COREx_SYSTIMER_TARGET0_INT_MAP_REG	21	
54	System Timer	SYSTIMER_TARGET1_INTR	COREx_SYSTIMER_TARGET1_INT_MAP_REG	22	
55	System Timer	SYSTIMER_TARGET2_INTR	COREx_SYSTIMER_TARGET2_INT_MAP_REG	23	
56	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_IN_CHO_INTR	COREx_AHB_PDMA_IN_CHO_INT_MAP_REG	24	
57	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_IN_CH1_INTR	COREx_AHB_PDMA_IN_CH1_INT_MAP_REG	25	
58	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_IN_CH2_INTR	COREx_AHB_PDMA_IN_CH2_INT_MAP_REG	26	
59	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_OUT_CHO_INTR	COREx_AHB_PDMA_OUT_CHO_INT_MAP_REG	27	
60	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_OUT_CH1_INTR	COREx_AHB_PDMA_OUT_CH1_INT_MAP_REG	28	
61	GDMA Controller (GDMA-AHB, GDMA-AXI)	AHB_PDMA_OUT_CH2_INTR	COREx_AHB_PDMA_OUT_CH2_INT_MAP_REG	29	
62	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_IN_CHO_INTR	COREx_AXI_PDMA_IN_CHO_INT_MAP_REG	30	
63	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_IN_CH1_INTR	COREx_AXI_PDMA_IN_CH1_INT_MAP_REG	31	

No.	Chapter	Interrupt Source	Interrupt Source Mapping Register	Interrupt Status Register	
				Bit	Name
64	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_IN_CH2_INTR	COREx_AXI_PDMA_IN_CH2_INT_MAP_REG	0	COREx_INTR_STATUS_REG_2_REG
65	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_OUT_CHO_INTR	COREx_AXI_PDMA_OUT_CHO_INT_MAP_REG	1	
66	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_OUT_CH1_INTR	COREx_AXI_PDMA_OUT_CH1_INT_MAP_REG	2	
67	GDMA Controller (GDMA-AHB, GDMA-AXI)	AXI_PDMA_OUT_CH2_INTR	COREx_AXI_PDMA_OUT_CH2_INT_MAP_REG	3	
68	RSA Accelerator (RSA)	RSA_INTR	COREx_RSA_INT_MAP_REG	4	
69	AES Accelerator (AES)	AES_INTR	COREx_AES_INT_MAP_REG	5	
70	SHA Accelerator (SHA)	SHA_INTR	COREx_SHA_INT_MAP_REG	6	
71	SHA Accelerator (SHA)	ECC_INTR	COREx_ECC_INT_MAP_REG	7	
72	ECDSA Digital Signature Peripheral (ECDSA_DS)	ECDSA_INTR	COREx_ECDSA_INT_MAP_REG	8	
73	Key Manager	KM_INTR	COREx_KM_INT_MAP_REG	9	
74	GPIO Matrix and IO MUX	GPIO_INTRO	COREx_GPIO_INTRO_MAP_REG	10	
75	GPIO Matrix and IO MUX	GPIO_INTR1	COREx_GPIO_INTR1_MAP_REG	11	
76	GPIO Matrix and IO MUX	GPIO_INTR2	COREx_GPIO_INTR2_MAP_REG	12	
77	GPIO Matrix and IO MUX	GPIO_INTR3	COREx_GPIO_INTR3_MAP_REG	13	
78	GPIO Matrix and IO MUX	GPIO_PAD_COMP_INTR	COREx_GPIO_PAD_COMP_INT_MAP_REG	14	
79	System Registers (SYSREG)	CPU_INTR_FROM_CPU_0	COREx_CPU_INTR_FROM_CPU_0_MAP_REG	15	
80	System Registers (SYSREG)	CPU_INTR_FROM_CPU_1	COREx_CPU_INTR_FROM_CPU_1_MAP_REG	16	
81	System Registers (SYSREG)	CPU_INTR_FROM_CPU_2	COREx_CPU_INTR_FROM_CPU_2_MAP_REG	17	
82	System Registers (SYSREG)	CPU_INTR_FROM_CPU_3	COREx_CPU_INTR_FROM_CPU_3_MAP_REG	18	
83	n/a	reserved	reserved	19	
84	SPI Controller (SPI)	FLASH_MSPI_INTR	COREx_FLASH_MSPI_INT_MAP_REG	20	
85	MIPI CSI	CSI_BRIDGE_INTR	COREx_CSI_BRIDGE_INT_MAP_REG	21	
86	MIPI DSI [to be added later]	DSI_BRIDGE_INTR	COREx_DSI_BRIDGE_INT_MAP_REG	22	
87	MIPI CSI	CSI_INTR	COREx_CSI_INT_MAP_REG	23	
88	MIPI DSI [to be added later]	DSI_INTR	COREx_DSI_INT_MAP_REG	24	
89	Ethernet Media Access Controller (EMAC)	GMII_PHY_INTR	COREx_GMII_PHY_INT_MAP_REG	25	
90	Ethernet Media Access Controller (EMAC)	LPI_INTR	COREx_LPI_INT_MAP_REG	26	
91	Ethernet Media Access Controller (EMAC)	PMT_INTR	COREx_PMT_INT_MAP_REG	27	
92	Ethernet Media Access Controller (EMAC)	ETH_MAC_INTR	COREx_ETH_MAC_INT_MAP_REG	28	
93	USB 2.0 High-Speed OTG	USB_OTG_INTR	COREx_USB_OTG_INT_MAP_REG	29	
94	USB 2.0 High-Speed OTG	USB_OTG_ENDP_MULTI_PROC_INTR	COREx_USB_OTG_ENDP_MULTI_PROC_INT_MAP_REG	30	
95	JPEG Codec	JPEG_INTR	COREx_JPEG_INT_MAP_REG	31	

No.	Chapter	Interrupt Source	Interrupt Source Mapping Register	Interrupt Status Register	
				Bit	Name
96	Pixel-Processing Accelerator (PPA)	PPA_INTR	COREx_PPA_INT_MAP_REG	0	COREx_INTR_STATUS_REG_3_REG
97	RISC-V Trace Encoder (TRACE)	CORE0_TRACE_INTR	COREx_CORE0_TRACE_INT_MAP_REG	1	
98	RISC-V Trace Encoder (TRACE)	CORE1_TRACE_INTR	COREx_CORE1_TRACE_INT_MAP_REG	2	
99	n/a	reserved	reserved	3	
100	Image Signal Processor (ISP)	ISP_INTR	COREx_ISP_INT_MAP_REG	4	
101	I3C Controller [to be added later]	I3C_MST_INTR	COREx_I3C_MST_INT_MAP_REG	5	
102	I3C Controller [to be added later]	I3C_SLV_INTR	COREx_I3C_SLV_INT_MAP_REG	6	
103	USB 2.0 Full-Speed OTG	USB_OTG11_INTR	COREx_USB_OTG11_INT_MAP_REG	7	
104	2D-DMA Controller (2D-DMA)	DMA2D_IN_CHO_INTR	COREx_DMA2D_IN_CHO_INT_MAP_REG	8	
105	2D-DMA Controller (2D-DMA)	DMA2D_IN_CH1_INTR	COREx_DMA2D_IN_CH1_INT_MAP_REG	9	
106	2D-DMA Controller (2D-DMA)	DMA2D_OUT_CHO_INTR	COREx_DMA2D_OUT_CHO_INT_MAP_REG	10	
107	2D-DMA Controller (2D-DMA)	DMA2D_OUT_CH1_INTR	COREx_DMA2D_OUT_CH1_INT_MAP_REG	11	
108	2D-DMA Controller (2D-DMA)	DMA2D_OUT_CH2_INTR	COREx_DMA2D_OUT_CH2_INT_MAP_REG	12	
109	SPI Controller (SPI)	PSRAM_MSPI_INTR	COREx_PSRAM_MSPI_INT_MAP_REG	13	
110	System Registers (SYSREG)	HP_SYSREG_INTR	COREx_HP_SYSREG_INT_MAP_REG	14	
111	Pulse Count Controller (PCNT)	PCNT_INTR	COREx_PCNT_INT_MAP_REG	15	
112	n/a	reserved	reserved	16	
113	Parallel IO Controller (PARLIO)	HP_PARLIO_RX_INTR	COREx_HP_PARLIO_RX_INT_MAP_REG	17	
114	Parallel IO Controller (PARLIO)	HP_PARLIO_TX_INTR	COREx_HP_PARLIO_TX_INT_MAP_REG	18	
115	H264 Encoder	H264_DMA_OUT_CHO_INTR	COREx_H264_DMA_OUT_CHO_INT_MAP_REG	19	
116	H264 Encoder	H264_DMA_OUT_CH1_INTR	COREx_H264_DMA_OUT_CH1_INT_MAP_REG	20	
117	H264 Encoder	H264_DMA_OUT_CH2_INTR	COREx_H264_DMA_OUT_CH2_INT_MAP_REG	21	
118	H264 Encoder	H264_DMA_OUT_CH3_INTR	COREx_H264_DMA_OUT_CH3_INT_MAP_REG	22	
119	H264 Encoder	H264_DMA_OUT_CH4_INTR	COREx_H264_DMA_OUT_CH4_INT_MAP_REG	23	
120	H264 Encoder	H264_DMA_IN_CHO_INTR	COREx_H264_DMA_IN_CHO_INT_MAP_REG	24	
121	H264 Encoder	H264_DMA_IN_CH1_INTR	COREx_H264_DMA_IN_CH1_INT_MAP_REG	25	
122	H264 Encoder	H264_DMA_IN_CH2_INTR	COREx_H264_DMA_IN_CH2_INT_MAP_REG	26	
123	H264 Encoder	H264_DMA_IN_CH3_INTR	COREx_H264_DMA_IN_CH3_INT_MAP_REG	27	
124	H264 Encoder	H264_DMA_IN_CH4_INTR	COREx_H264_DMA_IN_CH4_INT_MAP_REG	28	
125	H264 Encoder	H264_DMA_IN_CH5_INTR	COREx_H264_DMA_IN_CH5_INT_MAP_REG	29	
126	H264 Encoder	H264_REG_INTR	COREx_H264_REG_INT_MAP_REG	30	
127	Debug Assistant	ASSIST_DEBUG_INTR	COREx_ASSIST_DEBUG_INT_MAP_REG	31	

No.	Chapter	Interrupt Source	Interrupt Source Mapping Register	Interrupt Status Register	
				Bit	Name
128	2D-DMA Controller (2D-DMA)	DMA2D_IN_CH2_INTR	COREx_DMA2D_IN_CH2_INT_MAP_REG	0	COREx_INTR_STATUS_REG_4_REG
129	2D-DMA Controller (2D-DMA)	DMA2D_OUT_CH3_INTR	COREx_DMA2D_OUT_CH3_INT_MAP_REG	1	
130	AXI Performance Monitor [to be added later]	AXI_PERF_MON_INTR	COREx_AXI_PERF_MON_INT_MAP_REG	3	

12.4.2 Assign Peripheral Interrupt Source to HP CPU Interrupt

In this section, the following terms are used to describe the operation of the interrupt matrix.

- **SOURCE**: Stands for a peripheral interrupt source in Table 12.4-1.
- **CORE_x_SOURCE_MAP**: Stands for an interrupt source mapping register for the peripheral interrupt source (**SOURCE**) of HP CPU_x.
- Num_P: The index of HP CPU interrupts, which can be 16 ~ 47.
- Interrupt_P: Stands for the HP CPU interrupt numbered as Num_P.

12.4.2.1 Assign One Peripheral Interrupt Source (**SOURCE**) to HP CPU_x

Setting the corresponding source mapping register **CORE_x_SOURCE_MAP** of **SOURCE** to Num_P assigns this interrupt source to Interrupt_P of HP CPU_x.

12.4.2.2 Assign Multiple Peripheral Interrupt Sources (**SOURCE**) to HP CPU_x

Setting the corresponding source mapping register **CORE_x_SOURCE_MAP** of each interrupt source to the same Num_P assigns multiple sources to the same Interrupt_P of HP CPU_x. Any of these sources can trigger CPU Interrupt_P of HP CPU_x. In other words, the CPU Interrupt_P will be shared between these interrupt sources. When an interrupt signal is generated, HP CPU_x should check the interrupt status registers to determine which peripheral generated the interrupt. For more information, see Chapter 1 [High-Performance CPU](#).

12.4.2.3 Disable HP CPU_x Peripheral Interrupt Source (**SOURCE**)

Writing 0 to the **CORE_x_SOURCE_MAP** register disables the corresponding interrupt source.

12.4.3 Query Current Interrupt Status of HP CPU_x Peripheral Interrupt Source

After enabling peripheral interrupt sources, users can query the current interrupt status of HP CPU_x peripheral interrupt sources by reading the bit value in **CORE_x_INTR_STATUS_n** (read only). For the mapping between **CORE_x_INTR_STATUS_n** and peripheral interrupt sources, please refer to Table 12.4-1.

12.4.4 Interrupt Remapping

The RISC-V HP CPU_x (x can be 0 or 1) of ESP32-P4 supports Machine Mode and User Mode.

HP CPU_x provides a total of 32 peripheral interrupts. Each peripheral interrupt can be delegated by HP CPU_x as either a Machine-mode interrupt or a User-mode interrupt.

Under normal conditions, when HP CPU_x is running in Machine Mode, it can respond only to Machine-mode interrupts and cannot respond to User-mode interrupts. When HP CPU_x is running in User Mode, it can respond to both Machine-mode and User-mode interrupts.

To allow HP CPU_x to indirectly respond to User-mode interrupts while operating in Machine Mode, the system introduces the Interrupt Remap mechanism.

Interrupt remapping allows software to redirect HP CPU_x interrupts originally assigned to User Mode to a single specified Machine-mode interrupt. In this way, HP CPU_x can indirectly respond to User-mode interrupts while running in Machine Mode.

Interrupt remapping features:

- Enable interrupt remapping: Each peripheral interrupt source can be independently configured to participate in interrupt remapping. For example, setting `COREx_SOURCE_SRC_PASS_IN_SEC` to 1 enables interrupt remapping for interrupt matrix source `SOURCE`.
- Specify the target Machine-mode interrupt: All remapped User-mode interrupt signals are uniformly mapped to the same specified Machine-mode interrupt. For example, writing a value *n* to `COREx_INTR_SIG_IDX_ASSERT_IN_SEC_REG` selects the Machine-mode interrupt with interrupt number *n* as the remapping target.
- Access restrictions for remapping-related registers: the following registers and fields related to interrupt remapping are writable only when HP CPU_x is operating in Machine Mode:
 - `COREx_INTR_SIG_IDX_ASSERT_IN_SEC_REG`
 - `COREx_SOURCE_SRC_PASS_IN_SEC`
 - `COREx_SOURCE_SRC_IN_SEC_FLAG`
 - When `COREx_SOURCE_SRC_IN_SEC_FLAG` is set to 1, `COREx_SOURCE_MAP` is also writable only in Machine Mode.

When interrupt source `SOURCE` has been mapped to the peripheral interrupt signal *n* of CORE_x, interrupt remapping for source `SOURCE` occurs only when all of the following conditions are met:

- `COREx_SOURCE_SRC_PASS_IN_SEC` is set to 1
- CORE_x is currently operating in Machine Mode
- The peripheral interrupt signal *n* of CORE_x is internally delegated to User Mode by the CPU
- The target interrupt signal specified by `COREx_INTR_SIG_IDX_ASSERT_IN_SEC_REG` is internally delegated to Machine Mode by the CPU

12.5 Register Summary

The addresses in this section are relative to the interrupt matrix HP CPU0/HP CPU1 base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

12.5.1 HP CPU0 Interrupt Matrix Register Summary

Name	Description	Address	Access
COREO_LP_RTC_INT_MAP_REG	LP_RTC_INTR mapping register	0x0000	R/W
COREO_LP_WDT_INT_MAP_REG	LP_WDT_INTR mapping register	0x0004	R/W
COREO_LP_TIMER_REG_O_INT_MAP_REG	LP_TIMER_REG_O_INTR mapping register	0x0008	R/W
COREO_LP_TIMER_REG_1_INT_MAP_REG	LP_TIMER_REG_1_INTR mapping register	0x000C	R/W
COREO_MB_HP_INT_MAP_REG	MB_HP_INTR mapping register	0x0010	R/W
COREO_MB_LP_INT_MAP_REG	MB_LP_INTR mapping register	0x0014	R/W
COREO_PMU_REG_O_INT_MAP_REG	PMU_REG_O_INTR mapping register	0x0018	R/W
COREO_PMU_REG_1_INT_MAP_REG	PMU_REG_1_INTR mapping register	0x001C	R/W
COREO_LP_ANAPERI_INT_MAP_REG	LP_ANAPERI_INTR mapping register	0x0020	R/W
COREO_LP_ADC_INT_MAP_REG	LP_ADC_INTR mapping register	0x0024	R/W
COREO_LP_GPIO_INT_MAP_REG	LP_GPIO_INTR mapping register	0x0028	R/W
COREO_LP_I2C_INT_MAP_REG	LP_I2C_INTR mapping register	0x002C	R/W
COREO_LP_I2S_INT_MAP_REG	LP_I2S_INTR mapping register	0x0030	R/W
COREO_LP_SPI_INT_MAP_REG	LP_SPI_INTR mapping register	0x0034	R/W
COREO_LP_TOUCH_INT_MAP_REG	LP_TOUCH_INTR mapping register	0x0038	R/W
COREO_LP_TSENS_INT_MAP_REG	LP_TSENS_INTR mapping register	0x003C	R/W
COREO_LP_UART_INT_MAP_REG	LP_UART_INTR mapping register	0x0040	R/W
COREO_LP_EFUSE_INT_MAP_REG	LP_EFUSE_INTR mapping register	0x0044	R/W
COREO_LP_SW_INT_MAP_REG	LP_SW_INTR mapping register	0x0048	R/W
COREO_LP_SYSREG_INT_MAP_REG	LP_SYSREG_INTR mapping register	0x004C	R/W
COREO_LP_HUK_INT_MAP_REG	LP_HUK_INTR mapping register	0x0050	R/W

Name	Description	Address	Access
CORE0_SYS_ICM_INT_MAP_REG	SYS_ICM_INTR mapping register	0x0054	R/W
CORE0_USB_DEVICE_INT_MAP_REG	USB_DEVICE_INTR mapping register	0x0058	R/W
CORE0_SDIO_HOST_INT_MAP_REG	SDIO_HOST_INTR mapping register	0x005C	R/W
CORE0_GDMA_INT_MAP_REG	GDMA_INTR mapping register	0x0060	R/W
CORE0_SPI2_INT_MAP_REG	SPI2_INTR mapping register	0x0064	R/W
CORE0_SPI3_INT_MAP_REG	SPI3_INTR mapping register	0x0068	R/W
CORE0_I2SO_INT_MAP_REG	I2SO_INTR mapping register	0x006C	R/W
CORE0_I2S1_INT_MAP_REG	I2S1_INTR mapping register	0x0070	R/W
CORE0_I2S2_INT_MAP_REG	I2S2_INTR mapping register	0x0074	R/W
CORE0_UHCIO_INT_MAP_REG	UHCIO_INTR mapping register	0x0078	R/W
CORE0_UART0_INT_MAP_REG	UART0_INTR mapping register	0x007C	R/W
CORE0_UART1_INT_MAP_REG	UART1_INTR mapping register	0x0080	R/W
CORE0_UART2_INT_MAP_REG	UART2_INTR mapping register	0x0084	R/W
CORE0_UART3_INT_MAP_REG	UART3_INTR mapping register	0x0088	R/W
CORE0_UART4_INT_MAP_REG	UART4_INTR mapping register	0x008C	R/W
CORE0_LCD_CAM_INT_MAP_REG	LCD_CAM_INTR mapping register	0x0090	R/W
CORE0_ADC_INT_MAP_REG	ADC_INTR mapping register	0x0094	R/W
CORE0_PWM0_INT_MAP_REG	PWM0_INTR mapping register	0x0098	R/W
CORE0_PWM1_INT_MAP_REG	PWM1_INTR mapping register	0x009C	R/W
CORE0_TWAIO_INT_MAP_REG	TWAIO_INTR mapping register	0x00A0	R/W
CORE0_TWAI1_INT_MAP_REG	TWAI1_INTR mapping register	0x00A4	R/W
CORE0_TWAI2_INT_MAP_REG	TWAI2_INTR mapping register	0x00A8	R/W
CORE0_RMT_INT_MAP_REG	RMT_INTR mapping register	0x00AC	R/W
CORE0_I2CO_INT_MAP_REG	I2CO_INTR mapping register	0x00B0	R/W
CORE0_I2C1_INT_MAP_REG	I2C1_INTR mapping register	0x00B4	R/W
CORE0_TIMERGRPO_TO_INT_MAP_REG	TIMERGRPO_TO_INTR mapping register	0x00B8	R/W
CORE0_TIMERGRPO_T1_INT_MAP_REG	TIMERGRPO_T1_INTR mapping register	0x00BC	R/W
CORE0_TIMERGRPO_WDT_INT_MAP_REG	TIMERGRPO_WDT_INTR mapping register	0x00C0	R/W
CORE0_TIMERGRP1_TO_INT_MAP_REG	TIMERGRP1_TO_INTR mapping register	0x00C4	R/W

Name	Description	Address	Access
COREO_TIMERGRP1_T1_INT_MAP_REG	TIMERGRP1_T1_INTR mapping register	0x00C8	R/W
COREO_TIMERGRP1_WDT_INT_MAP_REG	TIMERGRP1_WDT_INTR mapping register	0x00CC	R/W
COREO_LEDC_INT_MAP_REG	LEDC_INTR mapping register	0x00D0	R/W
COREO_SYSTIMER_TARGET0_INT_MAP_REG	SYSTIMER_TARGET0_INTR mapping register	0x00D4	R/W
COREO_SYSTIMER_TARGET1_INT_MAP_REG	SYSTIMER_TARGET1_INTR mapping register	0x00D8	R/W
COREO_SYSTIMER_TARGET2_INT_MAP_REG	SYSTIMER_TARGET2_INTR mapping register	0x00DC	R/W
COREO_AHB_PDMA_IN_CHO_INT_MAP_REG	AHB_PDMA_IN_CHO_INTR mapping register	0x00E0	R/W
COREO_AHB_PDMA_IN_CH1_INT_MAP_REG	AHB_PDMA_IN_CH1_INTR mapping register	0x00E4	R/W
COREO_AHB_PDMA_IN_CH2_INT_MAP_REG	AHB_PDMA_IN_CH2_INTR mapping register	0x00E8	R/W
COREO_AHB_PDMA_OUT_CHO_INT_MAP_REG	AHB_PDMA_OUT_CHO_INTR mapping register	0x00EC	R/W
COREO_AHB_PDMA_OUT_CH1_INT_MAP_REG	AHB_PDMA_OUT_CH1_INTR mapping register	0x00F0	R/W
COREO_AHB_PDMA_OUT_CH2_INT_MAP_REG	AHB_PDMA_OUT_CH2_INTR mapping register	0x00F4	R/W
COREO_AXI_PDMA_IN_CHO_INT_MAP_REG	AXI_PDMA_IN_CHO_INTR mapping register	0x00F8	R/W
COREO_AXI_PDMA_IN_CH1_INT_MAP_REG	AXI_PDMA_IN_CH1_INTR mapping register	0x00FC	R/W
COREO_AXI_PDMA_IN_CH2_INT_MAP_REG	AXI_PDMA_IN_CH2_INTR mapping register	0x0100	R/W
COREO_AXI_PDMA_OUT_CHO_INT_MAP_REG	AXI_PDMA_OUT_CHO_INTR mapping register	0x0104	R/W
COREO_AXI_PDMA_OUT_CH1_INT_MAP_REG	AXI_PDMA_OUT_CH1_INTR mapping register	0x0108	R/W
COREO_AXI_PDMA_OUT_CH2_INT_MAP_REG	AXI_PDMA_OUT_CH2_INTR mapping register	0x010C	R/W
COREO_RSA_INT_MAP_REG	RSA_INTR mapping register	0x0110	R/W
COREO_AES_INT_MAP_REG	AES_INTR mapping register	0x0114	R/W
COREO_SHA_INT_MAP_REG	SHA_INTR mapping register	0x0118	R/W
COREO_ECC_INT_MAP_REG	ECC_INTR mapping register	0x011C	R/W
COREO_ECDSA_INT_MAP_REG	ECDSA_INTR mapping register	0x0120	R/W
COREO_KM_INT_MAP_REG	KM_INTR mapping register	0x0124	R/W
COREO_GPIO_INT0_MAP_REG	GPIO_INT0 mapping register	0x0128	R/W
COREO_GPIO_INT1_MAP_REG	GPIO_INTR1 mapping register	0x012C	R/W
COREO_GPIO_INT2_MAP_REG	GPIO_INTR2 mapping register	0x0130	R/W
COREO_GPIO_INT3_MAP_REG	GPIO_INTR3 mapping register	0x0134	R/W
COREO_GPIO_PAD_COMP_INT_MAP_REG	GPIO_PAD_COMP_INTR mapping register	0x0138	R/W

Name	Description	Address	Access
CORE0_CPU_INT_FROM_CPU_0_MAP_REG	CPU_INTR_FROM_CPU_0 mapping register	0x013C	R/W
CORE0_CPU_INT_FROM_CPU_1_MAP_REG	CPU_INTR_FROM_CPU_1 mapping register	0x0140	R/W
CORE0_CPU_INT_FROM_CPU_2_MAP_REG	CPU_INTR_FROM_CPU_2 mapping register	0x0144	R/W
CORE0_CPU_INT_FROM_CPU_3_MAP_REG	CPU_INTR_FROM_CPU_3 mapping register	0x0148	R/W
CORE0_FLASH_MSPI_INT_MAP_REG	FLASH_MSPI_INTR mapping register	0x0150	R/W
CORE0_CSI_BRIDGE_INT_MAP_REG	CSI_BRIDGE_INTR mapping register	0x0154	R/W
CORE0_DSI_BRIDGE_INT_MAP_REG	DSI_BRIDGE_INTR mapping register	0x0158	R/W
CORE0_CSI_INT_MAP_REG	CSI_INTR mapping register	0x015C	R/W
CORE0_DSI_INT_MAP_REG	DSI_INTR mapping register	0x0160	R/W
CORE0_GMII_PHY_INT_MAP_REG	GMII_PHY_INTR mapping register	0x0164	R/W
CORE0_LPI_INT_MAP_REG	LPI_INTR mapping register	0x0168	R/W
CORE0_PMT_INT_MAP_REG	PMT_INTR mapping register	0x016C	R/W
CORE0_ETH_MAC_INT_MAP_REG	ETH_MAC_INTR mapping register	0x0170	R/W
CORE0_USB_OTG_INT_MAP_REG	USB_OTG_INTR mapping register	0x0174	R/W
CORE0_USB_OTG_ENDP_MULTI_PROC_INT_MAP_REG	USB_OTG_ENDP_MULTI_PROC_INTR mapping register	0x0178	R/W
CORE0_JPEG_INT_MAP_REG	JPEG_INTR mapping register	0x017C	R/W
CORE0_PPA_INT_MAP_REG	PPA_INTR mapping register	0x0180	R/W
CORE0_CORE0_TRACE_INT_MAP_REG	CORE0_TRACE_INTR mapping register	0x0184	R/W
CORE0_CORE1_TRACE_INT_MAP_REG	CORE1_TRACE_INTR mapping register	0x0188	R/W
CORE0_ISP_INT_MAP_REG	ISP_INTR mapping register	0x0190	R/W
CORE0_I3C_MST_INT_MAP_REG	I3C_MST_INTR mapping register	0x0194	R/W
CORE0_I3C_SLV_INT_MAP_REG	I3C_SLV_INTR mapping register	0x0198	R/W
CORE0_USB_OTG11_INT_MAP_REG	USB_OTG11_INTR mapping register	0x019C	R/W
CORE0_DMA2D_IN_CHO_INT_MAP_REG	DMA2D_IN_CHO_INTR mapping register	0x01A0	R/W
CORE0_DMA2D_IN_CH1_INT_MAP_REG	DMA2D_IN_CH1_INTR mapping register	0x01A4	R/W
CORE0_DMA2D_OUT_CHO_INT_MAP_REG	DMA2D_OUT_CHO_INTR mapping register	0x01A8	R/W
CORE0_DMA2D_OUT_CH1_INT_MAP_REG	DMA2D_OUT_CH1_INTR mapping register	0x01AC	R/W
CORE0_DMA2D_OUT_CH2_INT_MAP_REG	DMA2D_OUT_CH2_INTR mapping register	0x01B0	R/W

Name	Description	Address	Access
CORE0_PSRAM_MSPI_INT_MAP_REG	PSRAM_MSPI_INTR mapping register	0x01B4	R/W
CORE0_HP_SYSREG_INT_MAP_REG	HP_SYSREG_INTR mapping register	0x01B8	R/W
CORE0_PCNT_INT_MAP_REG	PCNT_INTR mapping register	0x01BC	R/W
CORE0_HP_PARLIO_RX_INT_MAP_REG	HP_PARLIO_RX_INTR mapping register	0x01C4	R/W
CORE0_HP_PARLIO_TX_INT_MAP_REG	HP_PARLIO_TX_INTR mapping register	0x01C8	R/W
CORE0_H264_DMA2D_OUT_CHO_INT_MAP_REG	H264_DMA2D_OUT_CHO_INTR mapping register	0x01CC	R/W
CORE0_H264_DMA2D_OUT_CH1_INT_MAP_REG	H264_DMA2D_OUT_CH1_INTR mapping register	0x01D0	R/W
CORE0_H264_DMA2D_OUT_CH2_INT_MAP_REG	H264_DMA2D_OUT_CH2_INTR mapping register	0x01D4	R/W
CORE0_H264_DMA2D_OUT_CH3_INT_MAP_REG	H264_DMA2D_OUT_CH3_INTR mapping register	0x01D8	R/W
CORE0_H264_DMA2D_OUT_CH4_INT_MAP_REG	H264_DMA2D_OUT_CH4_INTR mapping register	0x01DC	R/W
CORE0_H264_DMA2D_IN_CHO_INT_MAP_REG	H264_DMA2D_IN_CHO_INTR mapping register	0x01E0	R/W
CORE0_H264_DMA2D_IN_CH1_INT_MAP_REG	H264_DMA2D_IN_CH1_INTR mapping register	0x01E4	R/W
CORE0_H264_DMA2D_IN_CH2_INT_MAP_REG	H264_DMA2D_IN_CH2_INTR mapping register	0x01E8	R/W
CORE0_H264_DMA2D_IN_CH3_INT_MAP_REG	H264_DMA2D_IN_CH3_INTR mapping register	0x01EC	R/W
CORE0_H264_DMA2D_IN_CH4_INT_MAP_REG	H264_DMA2D_IN_CH4_INTR mapping register	0x01F0	R/W
CORE0_H264_DMA2D_IN_CH5_INT_MAP_REG	H264_DMA2D_IN_CH5_INTR mapping register	0x01F4	R/W
CORE0_H264_REG_INT_MAP_REG	H264_REG_INTR mapping register	0x01F8	R/W
CORE0_ASSIST_DEBUG_INT_MAP_REG	ASSIST_DEBUG_INTR mapping register	0x01FC	R/W
CORE0_INTR_STATUS_REG_0_REG	Status register for interrupt sources 0 ~ 31	0x0200	RO
CORE0_INTR_STATUS_REG_1_REG	Status register for interrupt sources 32 ~ 63	0x0204	RO
CORE0_INTR_STATUS_REG_2_REG	Status register for interrupt sources 64 ~ 95	0x0208	RO
CORE0_INTR_STATUS_REG_3_REG	Status register for interrupt sources 96 ~ 127	0x020C	RO
CORE0_CLOCK_GATE_REG	Clock gating register	0x0210	R/W
CORE0_DMA2D_IN_CH2_INT_MAP_REG	DMA2D_IN_CH2_INTR mapping register	0x214	R/W

Name	Description	Address	Access
CORE0_DMA2D_OUT_CH3_INTR_MAP_REG	DMA2D_OUT_CH3_INTR mapping register	0x218	R/W
CORE0_AXI_PERF_MON_INTR_MAP_REG	AXI_PERF_MON_INTR mapping register	0x21C	R/W
CORE0_INTR_STATUS_REG_4_REG	Status register for interrupt sources 128 ~ 130	0x0220	RO
CORE0_INTR_SIG_IDX_ASSERT_IN_SEC_REG	Interrupt delegation configuration register for CORE0	0x0228	R/W
CORE0_INTR_SEC_STATUS_REG	Interrupt delegation status register for CORE0	0x022C	RO
CORE0_INTR_SRC_PAS_IN_SEC_STATUS_0_REG	Interrupt delegation status register for interrupt sources 0 ~ 31	0x0230	RO
CORE0_INTR_SRC_PAS_IN_SEC_STATUS_1_REG	Interrupt delegation status register for interrupt sources 32 ~ 63	0x0234	RO
CORE0_INTR_SRC_PAS_IN_SEC_STATUS_2_REG	Interrupt delegation status register for interrupt sources 64 ~ 95	0x0238	RO
CORE0_INTR_SRC_PAS_IN_SEC_STATUS_3_REG	Interrupt delegation status register for interrupt sources 96 ~ 127	0x023C	RO
CORE0_INTR_SRC_PAS_IN_SEC_STATUS_4_REG	Interrupt delegation status register for interrupt sources 128 ~ 130	0x0240	RO
CORE0_INTERRUPT_REG_DATE_REG	Version control register	0x03FC	R/W

12.5.2 HP CPU1 Interrupt Matrix Register Summary

Name	Description	Address	Access
CORE1_LP_RTC_INTR_MAP_REG	LP_RTC_INTR mapping register	0x0000	R/W
CORE1_LP_WDT_INTR_MAP_REG	LP_WDT_INTR mapping register	0x0004	R/W
CORE1_LP_TIMER_REG_O_INTR_MAP_REG	LP_TIMER_REG_O_INTR mapping register	0x0008	R/W
CORE1_LP_TIMER_REG_1_INTR_MAP_REG	LP_TIMER_REG_1_INTR mapping register	0x000C	R/W
CORE1_MB_HP_INTR_MAP_REG	MB_HP_INTR mapping register	0x0010	R/W
CORE1_MB_LP_INTR_MAP_REG	MB_LP_INTR mapping register	0x0014	R/W
CORE1_PMU_REG_O_INTR_MAP_REG	PMU_REG_O_INTR mapping register	0x0018	R/W

Name	Description	Address	Access
CORE1_PMU_REG_1_INT_MAP_REG	PMU_REG_1_INTR mapping register	0x001C	R/W
CORE1_LP_ANAPERI_INT_MAP_REG	LP_ANAPERI_INTR mapping register	0x0020	R/W
CORE1_LP_ADC_INT_MAP_REG	LP_ADC_INTR mapping register	0x0024	R/W
CORE1_LP_GPIO_INT_MAP_REG	LP_GPIO_INTR mapping register	0x0028	R/W
CORE1_LP_I2C_INT_MAP_REG	LP_I2C_INTR mapping register	0x002C	R/W
CORE1_LP_I2S_INT_MAP_REG	LP_I2S_INTR mapping register	0x0030	R/W
CORE1_LP_SPI_INT_MAP_REG	LP_SPI_INTR mapping register	0x0034	R/W
CORE1_LP_TOUCH_INT_MAP_REG	LP_TOUCH_INTR mapping register	0x0038	R/W
CORE1_LP_TSENS_INT_MAP_REG	LP_TSENS_INTR mapping register	0x003C	R/W
CORE1_LP_UART_INT_MAP_REG	LP_UART_INTR mapping register	0x0040	R/W
CORE1_LP_EFUSE_INT_MAP_REG	LP_EFUSE_INTR mapping register	0x0044	R/W
CORE1_LP_SW_INT_MAP_REG	LP_SW_INTR mapping register	0x0048	R/W
CORE1_LP_SYSREG_INT_MAP_REG	LP_SYSREG_INTR mapping register	0x004C	R/W
CORE1_LP_HUK_INT_MAP_REG	LP_HUK_INTR mapping register	0x0050	R/W
CORE1_SYS_ICM_INT_MAP_REG	SYS_ICM_INTR mapping register	0x0054	R/W
CORE1_USB_DEVICE_INT_MAP_REG	USB_DEVICE_INTR mapping register	0x0058	R/W
CORE1_SDIO_HOST_INT_MAP_REG	SDIO_HOST_INTR mapping register	0x005C	R/W
CORE1_GDMA_INT_MAP_REG	GDMA_INTR mapping register	0x0060	R/W
CORE1_SPI2_INT_MAP_REG	SPI2_INTR mapping register	0x0064	R/W
CORE1_SPI3_INT_MAP_REG	SPI3_INTR mapping register	0x0068	R/W
CORE1_I2S0_INT_MAP_REG	I2S0_INTR mapping register	0x006C	R/W
CORE1_I2S1_INT_MAP_REG	I2S1_INTR mapping register	0x0070	R/W
CORE1_I2S2_INT_MAP_REG	I2S2_INTR mapping register	0x0074	R/W
CORE1_UHCIO_INT_MAP_REG	UHCIO_INTR mapping register	0x0078	R/W
CORE1_UART0_INT_MAP_REG	UART0_INTR mapping register	0x007C	R/W
CORE1_UART1_INT_MAP_REG	UART1_INTR mapping register	0x0080	R/W
CORE1_UART2_INT_MAP_REG	UART2_INTR mapping register	0x0084	R/W
CORE1_UART3_INT_MAP_REG	UART3_INTR mapping register	0x0088	R/W
CORE1_UART4_INT_MAP_REG	UART4_INTR mapping register	0x008C	R/W

Name	Description	Address	Access
CORE1_LCD_CAM_INT_MAP_REG	LCD_CAM_INTR mapping register	0x0090	R/W
CORE1_ADC_INT_MAP_REG	ADC_INTR mapping register	0x0094	R/W
CORE1_PWMO_INT_MAP_REG	PWMO_INTR mapping register	0x0098	R/W
CORE1_PWM1_INT_MAP_REG	PWM1_INTR mapping register	0x009C	R/W
CORE1_TWAI0_INT_MAP_REG	TWAI0_INTR mapping register	0x00A0	R/W
CORE1_TWAI1_INT_MAP_REG	TWAI1_INTR mapping register	0x00A4	R/W
CORE1_TWAI2_INT_MAP_REG	TWAI2_INTR mapping register	0x00A8	R/W
CORE1_RMT_INT_MAP_REG	RMT_INTR mapping register	0x00AC	R/W
CORE1_I2CO_INT_MAP_REG	I2CO_INTR mapping register	0x00B0	R/W
CORE1_I2C1_INT_MAP_REG	I2C1_INTR mapping register	0x00B4	R/W
CORE1_TIMERGRPO_TO_INT_MAP_REG	TIMERGRPO_TO_INTR mapping register	0x00B8	R/W
CORE1_TIMERGRPO_T1_INT_MAP_REG	TIMERGRPO_T1_INTR mapping register	0x00BC	R/W
CORE1_TIMERGRPO_WDT_INT_MAP_REG	TIMERGRPO_WDT_INTR mapping register	0x00C0	R/W
CORE1_TIMERGRP1_TO_INT_MAP_REG	TIMERGRP1_TO_INTR mapping register	0x00C4	R/W
CORE1_TIMERGRP1_T1_INT_MAP_REG	TIMERGRP1_T1_INTR mapping register	0x00C8	R/W
CORE1_TIMERGRP1_WDT_INT_MAP_REG	TIMERGRP1_WDT_INTR mapping register	0x00CC	R/W
CORE1_LEDC_INT_MAP_REG	LEDC_INTR mapping register	0x00D0	R/W
CORE1_SYSTIMER_TARGET0_INT_MAP_REG	SYSTIMER_TARGET0_INTR mapping register	0x00D4	R/W
CORE1_SYSTIMER_TARGET1_INT_MAP_REG	SYSTIMER_TARGET1_INTR mapping register	0x00D8	R/W
CORE1_SYSTIMER_TARGET2_INT_MAP_REG	SYSTIMER_TARGET2_INTR mapping register	0x00DC	R/W
CORE1_AHB_PDMA_IN_CHO_INT_MAP_REG	AHB_PDMA_IN_CHO_INTR mapping register	0x00E0	R/W
CORE1_AHB_PDMA_IN_CH1_INT_MAP_REG	AHB_PDMA_IN_CH1_INTR mapping register	0x00E4	R/W
CORE1_AHB_PDMA_IN_CH2_INT_MAP_REG	AHB_PDMA_IN_CH2_INTR mapping register	0x00E8	R/W
CORE1_AHB_PDMA_OUT_CHO_INT_MAP_REG	AHB_PDMA_OUT_CHO_INTR mapping register	0x00EC	R/W
CORE1_AHB_PDMA_OUT_CH1_INT_MAP_REG	AHB_PDMA_OUT_CH1_INTR mapping register	0x00F0	R/W
CORE1_AHB_PDMA_OUT_CH2_INT_MAP_REG	AHB_PDMA_OUT_CH2_INTR mapping register	0x00F4	R/W
CORE1_AXI_PDMA_IN_CHO_INT_MAP_REG	AXI_PDMA_IN_CHO_INTR mapping register	0x00F8	R/W
CORE1_AXI_PDMA_IN_CH1_INT_MAP_REG	AXI_PDMA_IN_CH1_INTR mapping register	0x00FC	R/W
CORE1_AXI_PDMA_IN_CH2_INT_MAP_REG	AXI_PDMA_IN_CH2_INTR mapping register	0x0100	R/W

Name	Description	Address	Access
CORE1_AXI_PDMA_OUT_CHO_INT_MAP_REG	AXI_PDMA_OUT_CHO_INTR mapping register	0x0104	R/W
CORE1_AXI_PDMA_OUT_CH1_INT_MAP_REG	AXI_PDMA_OUT_CH1_INTR mapping register	0x0108	R/W
CORE1_AXI_PDMA_OUT_CH2_INT_MAP_REG	AXI_PDMA_OUT_CH2_INTR mapping register	0x010C	R/W
CORE1_RSA_INT_MAP_REG	RSA_INTR mapping register	0x0110	R/W
CORE1_AES_INT_MAP_REG	AES_INTR mapping register	0x0114	R/W
CORE1_SHA_INT_MAP_REG	SHA_INTR mapping register	0x0118	R/W
CORE1_ECC_INT_MAP_REG	ECC_INTR mapping register	0x011C	R/W
CORE1_ECDSA_INT_MAP_REG	ECDSA_INTR mapping register	0x0120	R/W
CORE1_KM_INT_MAP_REG	KM_INTR mapping register	0x0124	R/W
CORE1_GPIO_INT0_MAP_REG	GPIO_INT0 mapping register	0x0128	R/W
CORE1_GPIO_INT1_MAP_REG	GPIO_INTR1 mapping register	0x012C	R/W
CORE1_GPIO_INT2_MAP_REG	GPIO_INTR2 mapping register	0x0130	R/W
CORE1_GPIO_INT3_MAP_REG	GPIO_INTR3 mapping register	0x0134	R/W
CORE1_GPIO_PAD_COMP_INT_MAP_REG	GPIO_PAD_COMP_INTR mapping register	0x0138	R/W
CORE1_CPU_INT_FROM_CPU_0_MAP_REG	CPU_INTR_FROM_CPU_0 mapping register	0x013C	R/W
CORE1_CPU_INT_FROM_CPU_1_MAP_REG	CPU_INTR_FROM_CPU_1 mapping register	0x0140	R/W
CORE1_CPU_INT_FROM_CPU_2_MAP_REG	CPU_INTR_FROM_CPU_2 mapping register	0x0144	R/W
CORE1_CPU_INT_FROM_CPU_3_MAP_REG	CPU_INTR_FROM_CPU_3 mapping register	0x0148	R/W
CORE1_FLASH_MSPI_INT_MAP_REG	FLASH_MSPI_INTR mapping register	0x0150	R/W
CORE1_CSI_BRIDGE_INT_MAP_REG	CSI_BRIDGE_INTR mapping register	0x0154	R/W
CORE1_DSI_BRIDGE_INT_MAP_REG	DSI_BRIDGE_INTR mapping register	0x0158	R/W
CORE1_CSI_INT_MAP_REG	CSI_INTR mapping register	0x015C	R/W
CORE1_DSI_INT_MAP_REG	DSI_INTR mapping register	0x0160	R/W
CORE1_GMII_PHY_INT_MAP_REG	GMII_PHY_INTR mapping register	0x0164	R/W
CORE1_LPI_INT_MAP_REG	LPI_INTR mapping register	0x0168	R/W
CORE1_PMT_INT_MAP_REG	PMT_INTR mapping register	0x016C	R/W
CORE1_ETH_MAC_INT_MAP_REG	ETH_MAC_INTR mapping register	0x0170	R/W
CORE1_USB_OTG_INT_MAP_REG	USB_OTG_INTR mapping register	0x0174	R/W

Name	Description	Address	Access
CORE1_USB_OTG_ENDP_MULTI_PROC_INT_MAP_REG	USB_OTG_ENDP_MULTI_PROC_INTR mapping register	0x0178	R/W
CORE1_JPEG_INT_MAP_REG	JPEG_INTR mapping register	0x017C	R/W
CORE1_PPA_INT_MAP_REG	PPA_INTR mapping register	0x0180	R/W
CORE1_CORE0_TRACE_INT_MAP_REG	CORE0_TRACE_INTR mapping register	0x0184	R/W
CORE1_CORE1_TRACE_INT_MAP_REG	CORE1_TRACE_INTR mapping register	0x0188	R/W
CORE1_ISP_INT_MAP_REG	ISP_INTR mapping register	0x0190	R/W
CORE1_I3C_MST_INT_MAP_REG	I3C_MST_INTR mapping register	0x0194	R/W
CORE1_I3C_SLV_INT_MAP_REG	I3C_SLV_INTR mapping register	0x0198	R/W
CORE1_USB_OTG11_INT_MAP_REG	USB_OTG11_INTR mapping register	0x019C	R/W
CORE1_DMA2D_IN_CHO_INT_MAP_REG	DMA2D_IN_CHO_INTR mapping register	0x01A0	R/W
CORE1_DMA2D_IN_CH1_INT_MAP_REG	DMA2D_IN_CH1_INTR mapping register	0x01A4	R/W
CORE1_DMA2D_OUT_CHO_INT_MAP_REG	DMA2D_OUT_CHO_INTR mapping register	0x01A8	R/W
CORE1_DMA2D_OUT_CH1_INT_MAP_REG	DMA2D_OUT_CH1_INTR mapping register	0x01AC	R/W
CORE1_DMA2D_OUT_CH2_INT_MAP_REG	DMA2D_OUT_CH2_INTR mapping register	0x01B0	R/W
CORE1_PSRAM_MSPI_INT_MAP_REG	PSRAM_MSPI_INTR mapping register	0x01B4	R/W
CORE1_HP_SYSREG_INT_MAP_REG	HP_SYSREG_INTR mapping register	0x01B8	R/W
CORE1_PCNT_INT_MAP_REG	PCNT_INTR mapping register	0x01BC	R/W
CORE1_HP_PARLIO_RX_INT_MAP_REG	HP_PARLIO_RX_INTR mapping register	0x01C4	R/W
CORE1_HP_PARLIO_TX_INT_MAP_REG	HP_PARLIO_TX_INTR mapping register	0x01C8	R/W
CORE1_H264_DMA2D_OUT_CHO_INT_MAP_REG	H264_DMA2D_OUT_CHO_INTR mapping register	0x01CC	R/W
CORE1_H264_DMA2D_OUT_CH1_INT_MAP_REG	H264_DMA2D_OUT_CH1_INTR mapping register	0x01D0	R/W
CORE1_H264_DMA2D_OUT_CH2_INT_MAP_REG	H264_DMA2D_OUT_CH2_INTR mapping register	0x01D4	R/W
CORE1_H264_DMA2D_OUT_CH3_INT_MAP_REG	H264_DMA2D_OUT_CH3_INTR mapping register	0x01D8	R/W

Name	Description	Address	Access
CORE1_H264_DMA2D_OUT_CH4_INT_MAP_REG	H264_DMA2D_OUT_CH4_INTR mapping register	0x01DC	R/W
CORE1_H264_DMA2D_IN_CHO_INT_MAP_REG	H264_DMA2D_IN_CHO_INTR mapping register	0x01E0	R/W
CORE1_H264_DMA2D_IN_CH1_INT_MAP_REG	H264_DMA2D_IN_CH1_INTR mapping register	0x01E4	R/W
CORE1_H264_DMA2D_IN_CH2_INT_MAP_REG	H264_DMA2D_IN_CH2_INTR mapping register	0x01E8	R/W
CORE1_H264_DMA2D_IN_CH3_INT_MAP_REG	H264_DMA2D_IN_CH3_INTR mapping register	0x01EC	R/W
CORE1_H264_DMA2D_IN_CH4_INT_MAP_REG	H264_DMA2D_IN_CH4_INTR mapping register	0x01F0	R/W
CORE1_H264_DMA2D_IN_CH5_INT_MAP_REG	H264_DMA2D_IN_CH5_INTR mapping register	0x01F4	R/W
CORE1_H264_REG_INT_MAP_REG	H264_REG_INTR mapping register	0x01F8	R/W
CORE1_ASSIST_DEBUG_INT_MAP_REG	ASSIST_DEBUG_INTR mapping register	0x01FC	R/W
CORE1_INTR_STATUS_REG_0_REG	Status register for interrupt sources 0 ~ 31	0x0200	RO
CORE1_INTR_STATUS_REG_1_REG	Status register for interrupt sources 32 ~ 63	0x0204	RO
CORE1_INTR_STATUS_REG_2_REG	Status register for interrupt sources 64 ~ 95	0x0208	RO
CORE1_INTR_STATUS_REG_3_REG	Status register for interrupt sources 96 ~ 127	0x020C	RO
CORE1_CLOCK_GATE_REG	Clock gating register	0x0210	R/W
CORE1_DMA2D_IN_CH2_INT_MAP_REG	DMA2D_IN_CH2_INTR mapping register	0x214	R/W
CORE1_DMA2D_OUT_CH3_INT_MAP_REG	DMA2D_OUT_CH3_INTR mapping register	0x218	R/W
CORE1_AXI_PERF_MON_INT_MAP_REG	AXI_PERF_MON_INTR mapping register	0x21C	R/W
CORE1_INTR_STATUS_REG_4_REG	Status register for interrupt sources 128 ~ 130	0x0220	RO
CORE1_INTR_SIG_IDX_ASSERT_IN_SEC_REG	CORE1 interrupt delegation configuration register	0x0228	R/W
CORE1_INTR_SEC_STATUS_REG	CORE1 interrupt delegation status register	0x022C	RO
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_0_REG	Interrupt delegation status register for interrupt sources 0 ~ 31	0x0230	RO
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_1_REG	Interrupt delegation status register for interrupt sources 32 ~ 63	0x0234	RO
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_2_REG	Interrupt delegation status register for interrupt sources 64 ~ 95	0x0238	RO

Name	Description	Address	Access
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_3_REG	Interrupt delegation status register for interrupt sources 96 ~ 127	0x023C	RO
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_4_REG	Interrupt delegation status register for interrupt sources 128 ~ 130	0x0240	RO
CORE1_INTERRUPT_REG_DATE_REG	Version control register	0x03FC	R/W

12.6 Registers

The addresses in this section are relative to the interrupt matrix base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

12.6.1 HP CPU0 Interrupt Matrix Registers

Register 12.1. COREO_LP_RTC_INT_MAP_REG (0x0000)

Register 12.2. COREO_LP_WDT_INT_MAP_REG (0x0004)

Register 12.3. COREO_LP_TIMER_REG_0_INT_MAP_REG (0x0008)

Register 12.4. COREO_LP_TIMER_REG_1_INT_MAP_REG (0x000C)

Register 12.5. COREO_MB_HP_INT_MAP_REG (0x0010)

Register 12.6. COREO_MB_LP_INT_MAP_REG (0x0014)

Register 12.7. COREO_PMU_REG_0_INT_MAP_REG (0x0018)

Register 12.8. COREO_PMU_REG_1_INT_MAP_REG (0x001C)

Register 12.9. COREO_LP_ANAPERI_INT_MAP_REG (0x0020)

Register 12.10. COREO_LP_ADC_INT_MAP_REG (0x0024)

Register 12.11. COREO_LP_GPIO_INT_MAP_REG (0x0028)

Register 12.12. COREO_LP_I2C_INT_MAP_REG (0x002C)

Register 12.13. COREO_LP_I2S_INT_MAP_REG (0x0030)

Register 12.14. COREO_LP_SPI_INT_MAP_REG (0x0034)

Register 12.15. COREO_LP_TOUCH_INT_MAP_REG (0x0038)

Register 12.16. COREO_LP_TSENS_INT_MAP_REG (0x003C)

Register 12.17. COREO_LP_UART_INT_MAP_REG (0x0040)

Register 12.18. COREO_LP_EFUSE_INT_MAP_REG (0x0044)

Register 12.19. COREO_LP_SW_INT_MAP_REG (0x0048)

Register 12.20. COREO_LP_SYSREG_INT_MAP_REG (0x004C)

Register 12.21. COREO_LP_HUK_INT_MAP_REG (0x0050)

Register 12.22. COREO_SYS_ICM_INT_MAP_REG (0x0054)

Register 12.23. COREO_USB_DEVICE_INT_MAP_REG (0x0058)

Register 12.24. COREO_SDIO_HOST_INT_MAP_REG (0x005C)

Register 12.25. COREO_GDMA_INT_MAP_REG (0x0060)

Register 12.26. COREO_SPI2_INT_MAP_REG (0x0064)

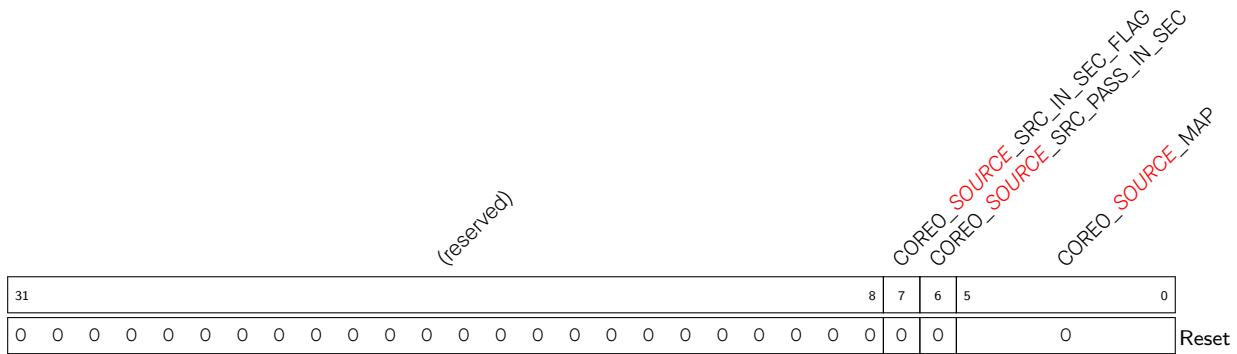
Register 12.27. COREO_SPI3_INT_MAP_REG (0x0068)

Register 12.28. COREO_I2SO_INT_MAP_REG (0x006C)

- Register 12.29. COREO_*I2S1_INT*_MAP_REG (0x0070)
- Register 12.30. COREO_*I2S2_INT*_MAP_REG (0x0074)
- Register 12.31. COREO_*UHCIO_INT*_MAP_REG (0x0078)
- Register 12.32. COREO_*UART0_INT*_MAP_REG (0x007C)
- Register 12.33. COREO_*UART1_INT*_MAP_REG (0x0080)
- Register 12.34. COREO_*UART2_INT*_MAP_REG (0x0084)
- Register 12.35. COREO_*UART3_INT*_MAP_REG (0x0088)
- Register 12.36. COREO_*UART4_INT*_MAP_REG (0x008C)
- Register 12.37. COREO_*LCD_CAM_INT*_MAP_REG (0x0090)
- Register 12.38. COREO_*ADC_INT*_MAP_REG (0x0094)
- Register 12.39. COREO_*PWMO_INT*_MAP_REG (0x0098)
- Register 12.40. COREO_*PWM1_INT*_MAP_REG (0x009C)
- Register 12.41. COREO_*TWAI0_INT*_MAP_REG (0x00A0)
- Register 12.42. COREO_*TWAI1_INT*_MAP_REG (0x00A4)
- Register 12.43. COREO_*TWAI2_INT*_MAP_REG (0x00A8)
- Register 12.44. COREO_*RMT_INT*_MAP_REG (0x00AC)
- Register 12.45. COREO_*I2CO_INT*_MAP_REG (0x00B0)
- Register 12.46. COREO_*I2C1_INT*_MAP_REG (0x00B4)
- Register 12.47. COREO_*TIMERGRPO_TO_INT*_MAP_REG (0x00B8)
- Register 12.48. COREO_*TIMERGRPO_T1_INT*_MAP_REG (0x00BC)
- Register 12.49. COREO_*TIMERGRPO_WDT_INT*_MAP_REG (0x00C0)
- Register 12.50. COREO_*TIMERGRP1_TO_INT*_MAP_REG (0x00C4)
- Register 12.51. COREO_*TIMERGRP1_T1_INT*_MAP_REG (0x00C8)
- Register 12.52. COREO_*TIMERGRP1_WDT_INT*_MAP_REG (0x00CC)
- Register 12.53. COREO_*LEDC_INT*_MAP_REG (0x00D0)
- Register 12.54. COREO_*SYSTIMER_TARGET0_INT*_MAP_REG (0x00D4)
- Register 12.55. COREO_*SYSTIMER_TARGET1_INT*_MAP_REG (0x00D8)
- Register 12.56. COREO_*SYSTIMER_TARGET2_INT*_MAP_REG (0x00DC)
- Register 12.57. COREO_*AHB_PDMA_IN_CHO_INT*_MAP_REG (0x00E0)
- Register 12.58. COREO_*AHB_PDMA_IN_CH1_INT*_MAP_REG (0x00E4)
- Register 12.59. COREO_*AHB_PDMA_IN_CH2_INT*_MAP_REG (0x00E8)
- Register 12.60. COREO_*AHB_PDMA_OUT_CHO_INT*_MAP_REG (0x00EC)
- Register 12.61. COREO_*AHB_PDMA_OUT_CH1_INT*_MAP_REG (0x00F0)
- Register 12.62. COREO_*AHB_PDMA_OUT_CH2_INT*_MAP_REG (0x00F4)
- Register 12.63. COREO_*AXI_PDMA_IN_CHO_INT*_MAP_REG (0x00F8)

- Register 12.64. CORE0_ [AXI_PDMA_IN_CH1_INT_MAP_REG](#) (0x00FC)
- Register 12.65. CORE0_ [AXI_PDMA_IN_CH2_INT_MAP_REG](#) (0x0100)
- Register 12.66. CORE0_ [AXI_PDMA_OUT_CH0_INT_MAP_REG](#) (0x0104)
- Register 12.67. CORE0_ [AXI_PDMA_OUT_CH1_INT_MAP_REG](#) (0x0108)
- Register 12.68. CORE0_ [AXI_PDMA_OUT_CH2_INT_MAP_REG](#) (0x010C)
- Register 12.69. CORE0_ [RSA_INT_MAP_REG](#) (0x0110)
- Register 12.70. CORE0_ [AES_INT_MAP_REG](#) (0x0114)
- Register 12.71. CORE0_ [SHA_INT_MAP_REG](#) (0x0118)
- Register 12.72. CORE0_ [ECC_INT_MAP_REG](#) (0x011C)
- Register 12.73. CORE0_ [ECDSA_INT_MAP_REG](#) (0x0120)
- Register 12.74. CORE0_ [KM_INT_MAP_REG](#) (0x0124)
- Register 12.75. CORE0_ [GPIO_INT0_MAP_REG](#) (0x0128)
- Register 12.76. CORE0_ [GPIO_INT1_MAP_REG](#) (0x012C)
- Register 12.77. CORE0_ [GPIO_INT2_MAP_REG](#) (0x0130)
- Register 12.78. CORE0_ [GPIO_INT3_MAP_REG](#) (0x0134)
- Register 12.79. CORE0_ [GPIO_PAD_COMP_INT_MAP_REG](#) (0x0138)
- Register 12.80. CORE0_ [CPU_INT_FROM_CPU_0_MAP_REG](#) (0x013C)
- Register 12.81. CORE0_ [CPU_INT_FROM_CPU_1_MAP_REG](#) (0x0140)
- Register 12.82. CORE0_ [CPU_INT_FROM_CPU_2_MAP_REG](#) (0x0144)
- Register 12.83. CORE0_ [CPU_INT_FROM_CPU_3_MAP_REG](#) (0x0148)
- Register 12.84. CORE0_ [FLASH_MSPI_INT_MAP_REG](#) (0x0150)
- Register 12.85. CORE0_ [CSI_BRIDGE_INT_MAP_REG](#) (0x0154)
- Register 12.86. CORE0_ [DSI_BRIDGE_INT_MAP_REG](#) (0x0158)
- Register 12.87. CORE0_ [CSI_INT_MAP_REG](#) (0x015C)
- Register 12.88. CORE0_ [DSI_INT_MAP_REG](#) (0x0160)
- Register 12.89. CORE0_ [GMII_PHY_INT_MAP_REG](#) (0x0164)
- Register 12.90. CORE0_ [LPI_INT_MAP_REG](#) (0x0168)
- Register 12.91. CORE0_ [PMT_INT_MAP_REG](#) (0x016C)
- Register 12.92. CORE0_ [ETH_MAC_INT_MAP_REG](#) (0x0170)
- Register 12.93. CORE0_ [USB_OTG_INT_MAP_REG](#) (0x0174)
- Register 12.94. CORE0_ [USB_OTG_ENDP_MULTI_PROC_INT_MAP_REG](#) (0x0178)
- Register 12.95. CORE0_ [JPEG_INT_MAP_REG](#) (0x017C)
- Register 12.96. CORE0_ [PPA_INT_MAP_REG](#) (0x0180)
- Register 12.97. CORE0_ [CORE0_TRACE_INT_MAP_REG](#) (0x0184)
- Register 12.98. CORE0_ [CORE1_TRACE_INT_MAP_REG](#) (0x0188)

- Register 12.99. CORE0_*ISP_INT*_MAP_REG (0x0190)
- Register 12.100. CORE0_*I3C_MST_INT*_MAP_REG (0x0194)
- Register 12.101. CORE0_*I3C_SLV_INT*_MAP_REG (0x0198)
- Register 12.102. CORE0_*USB_OTG11_INT*_MAP_REG (0x019C)
- Register 12.103. CORE0_*DMA2D_IN_CHO_INT*_MAP_REG (0x01A0)
- Register 12.104. CORE0_*DMA2D_IN_CH1_INT*_MAP_REG (0x01A4)
- Register 12.105. CORE0_*DMA2D_OUT_CHO_INT*_MAP_REG (0x01A8)
- Register 12.106. CORE0_*DMA2D_OUT_CH1_INT*_MAP_REG (0x01AC)
- Register 12.107. CORE0_*DMA2D_OUT_CH2_INT*_MAP_REG (0x01B0)
- Register 12.108. CORE0_*PSRAM_MSPI_INT*_MAP_REG (0x01B4)
- Register 12.109. CORE0_*HP_SYSREG_INT*_MAP_REG (0x01B8)
- Register 12.110. CORE0_*PCNT_INT*_MAP_REG (0x01BC)
- Register 12.111. CORE0_*HP_PARLIO_RX_INT*_MAP_REG (0x01C4)
- Register 12.112. CORE0_*HP_PARLIO_TX_INT*_MAP_REG (0x01C8)
- Register 12.113. CORE0_*H264_DMA2D_OUT_CHO_INT*_MAP_REG (0x01CC)
- Register 12.114. CORE0_*H264_DMA2D_OUT_CH1_INT*_MAP_REG (0x01D0)
- Register 12.115. CORE0_*H264_DMA2D_OUT_CH2_INT*_MAP_REG (0x01D4)
- Register 12.116. CORE0_*H264_DMA2D_OUT_CH3_INT*_MAP_REG (0x01D8)
- Register 12.117. CORE0_*H264_DMA2D_OUT_CH4_INT*_MAP_REG (0x01DC)
- Register 12.118. CORE0_*H264_DMA2D_IN_CHO_INT*_MAP_REG (0x01E0)
- Register 12.119. CORE0_*H264_DMA2D_IN_CH1_INT*_MAP_REG (0x01E4)
- Register 12.120. CORE0_*H264_DMA2D_IN_CH2_INT*_MAP_REG (0x01E8)
- Register 12.121. CORE0_*H264_DMA2D_IN_CH3_INT*_MAP_REG (0x01EC)
- Register 12.122. CORE0_*H264_DMA2D_IN_CH4_INT*_MAP_REG (0x01F0)
- Register 12.123. CORE0_*H264_DMA2D_IN_CH5_INT*_MAP_REG (0x01F4)
- Register 12.124. CORE0_*H264_REG_INT*_MAP_REG (0x01F8)
- Register 12.125. CORE0_*ASSIST_DEBUG_INT*_MAP_REG (0x01FC)
- Register 12.126. CORE0_*DMA2D_IN_CH2_INT*_MAP_REG (0x0214)
- Register 12.127. CORE0_*DMA2D_OUT_CH3_INT*_MAP_REG (0x0218)
- Register 12.128. CORE0_*AXI_PERF_MON_INT*_MAP_REG (0x021C)

Register 12.129. COREO_*SOURCE*_MAP_REG (0x0000 - 0x01FC, 0x0214 - 0x021C)

COREO_*SOURCE*_MAP When **COREO_*SOURCE*_SRC_IN_SEC_FLAG** is set to 1, this register can only be written by CPU while in Machine Mode.

Configure this register to map the interrupt signal from the peripheral interrupt sources *Source* to one of CPU0's peripheral interrupts. The valid configuration range is 16 ~ 47.

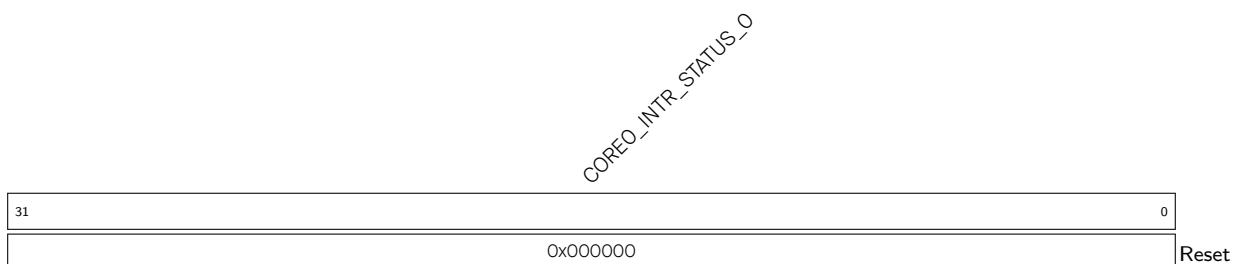
When set to 0 ~ 15, the corresponding peripheral interrupt source is disabled (recommended value: 0).

Other values are invalid.

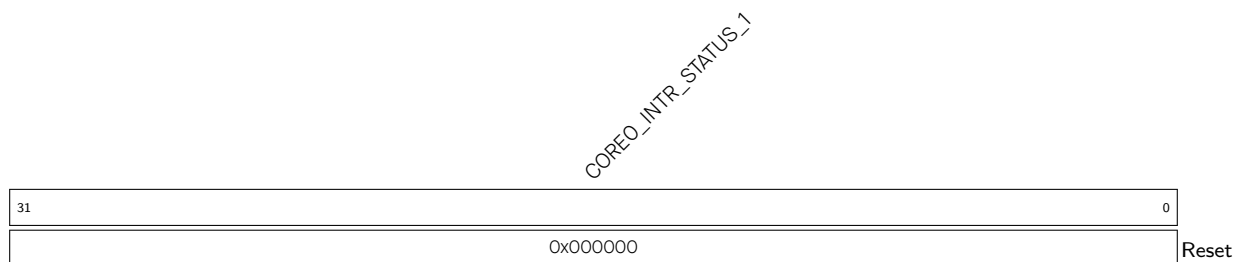
For the list of interrupt sources *Source*, see Table 12.4-1. (R/W)

COREO_*SOURCE*_SRC_PASS_IN_SEC Write 1 to allow the interrupt signal from interrupt source *Source* to be delegated to CPU0's Machine Mode interrupt. Writable only by the CPU in Machine Mode. (R/W)

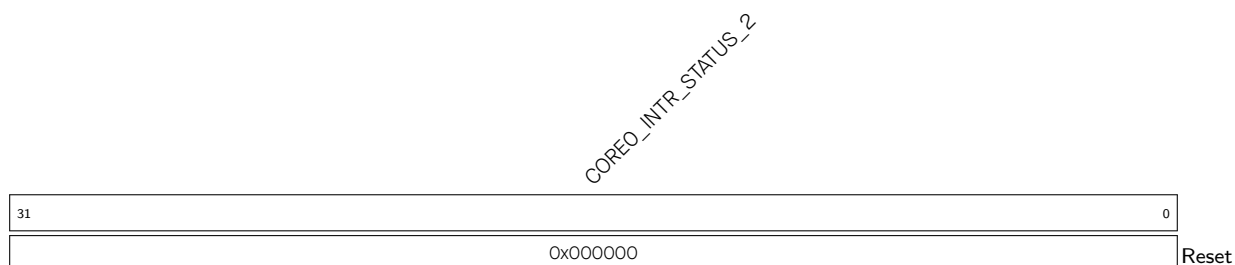
COREO_*SOURCE*_SRC_IN_SEC_FLAG Write 1 to configure the interrupt signal from interrupt source *Source* as a secure interrupt source (that is, its interrupt mapping can be configured only by the CPU running in Machine Mode). Writable only by the CPU in Machine Mode. (R/W)

Register 12.130. COREO_INTR_STATUS_REG_O_REG (0x0200)

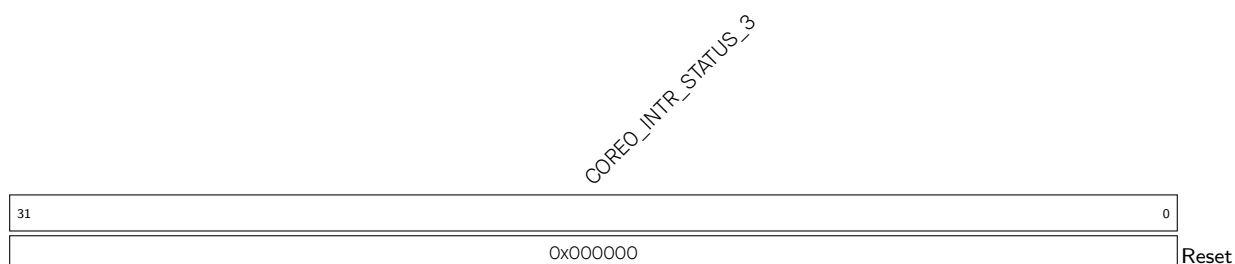
COREO_INTR_STATUS_O Represents the status of the interrupt sources from 0 ~ 31. Each bit corresponds to one interrupt source. (RO)

Register 12.131. COREO_INTR_STATUS_REG_1_REG (0x0204)

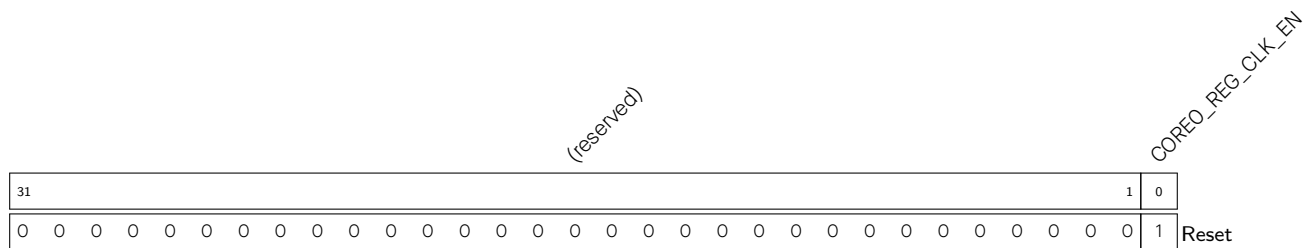
COREO_INTR_STATUS_1 Represents the status of the interrupt sources from 32 ~ 63. Each bit corresponds to one interrupt source. (RO)

Register 12.132. COREO_INTR_STATUS_REG_2_REG (0x0208)

COREO_INTR_STATUS_2 Represents the status of the interrupt sources from 64 ~ 95. Each bit corresponds to one interrupt source. (RO)

Register 12.133. COREO_INTR_STATUS_REG_3_REG (0x020C)

COREO_INTR_STATUS_3 Represents the status of the interrupt sources numbered from 96 ~ 127 (99 and 112 are reserved). Each bit corresponds to one interrupt source. (RO)

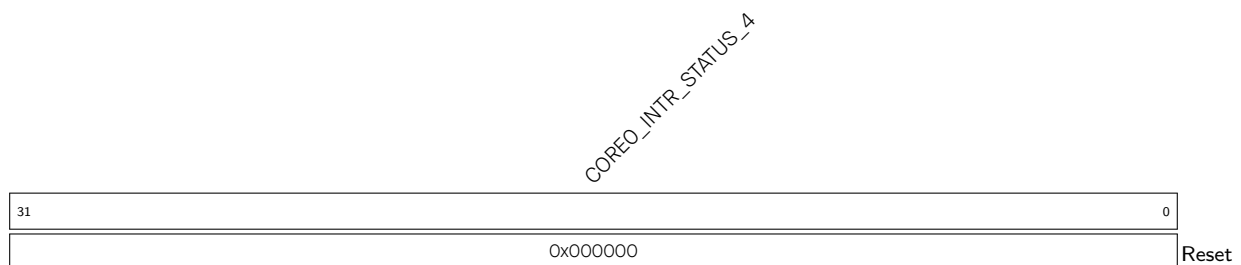
Register 12.134. COREO_CLOCK_GATE_REG (0x0210)

COREO_REG_CLK_EN Configures whether to force interrupt register clock-gate on.

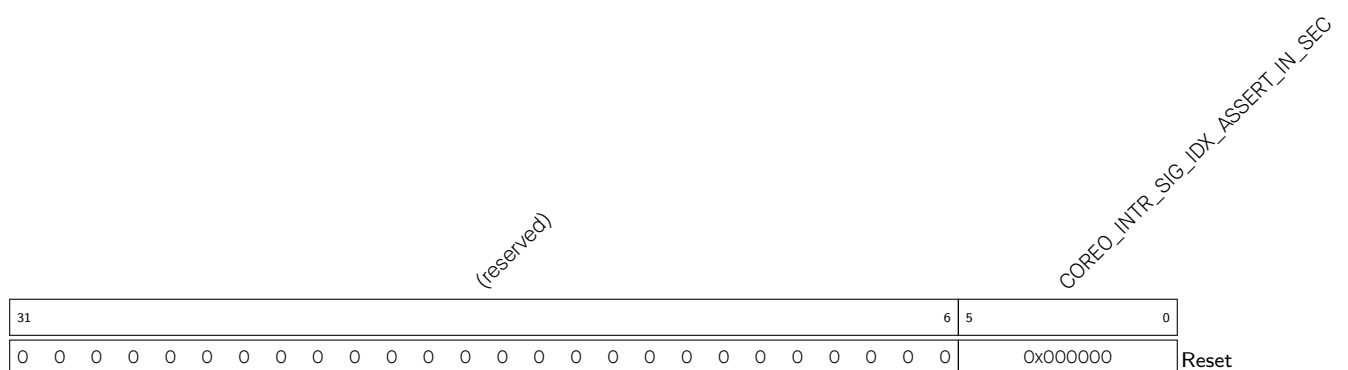
0: No effect

1: Force interrupt register clock-gate on

(R/W)

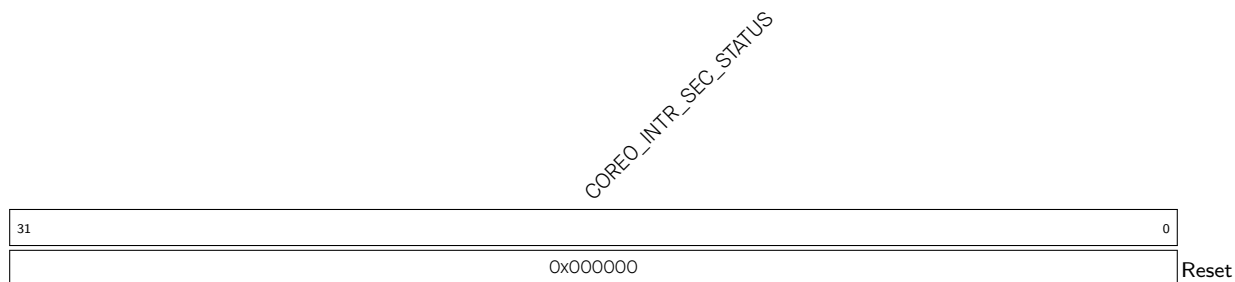
Register 12.135. COREO_INTR_STATUS_REG_4_REG (0x0220)

COREO_INTR_STATUS_4 Represents the status of the interrupt sources from 128 ~ 130. Each bit corresponds to one interrupt source. (RO)

Register 12.136. COREO_INTR_SIG_IDX_ASSERT_IN_SEC_REG (0x0228)

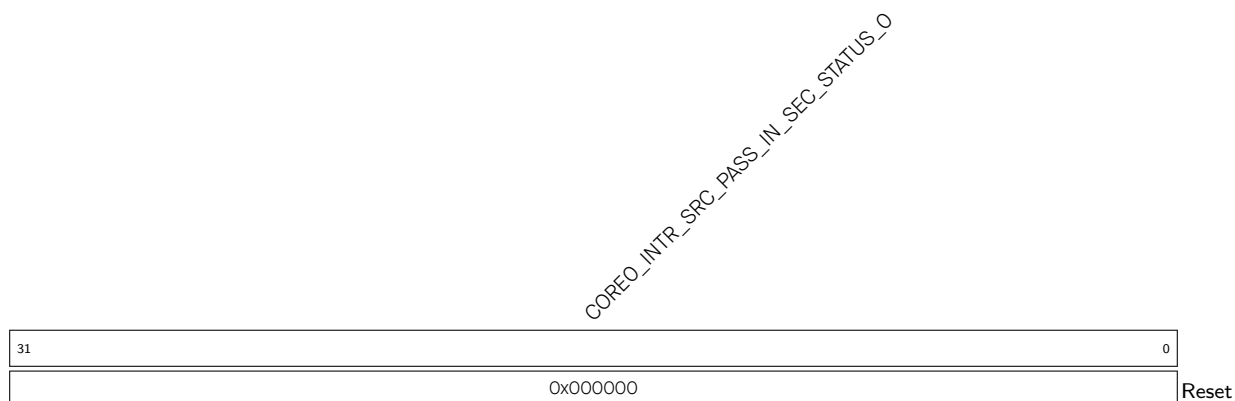
COREO_INTR_SIG_IDX_ASSERT_IN_SEC Configures which peripheral machine mode interrupt of CPU0 is used as the target for the interrupt delegation function. Valid values range from 16 ~ 47. (R/W)

Register 12.137. COREO_INTR_SEC_STATUS_REG (0x022C)



COREO_INTR_SEC_STATUS Represents status of the user mode interrupt signals that have been delegated to the machine mode interrupt when COREO is in machine mode. (RO)

Register 12.138. COREO_INTR_SRC_PASS_IN_SEC_STATUS_0_REG (0x0230)

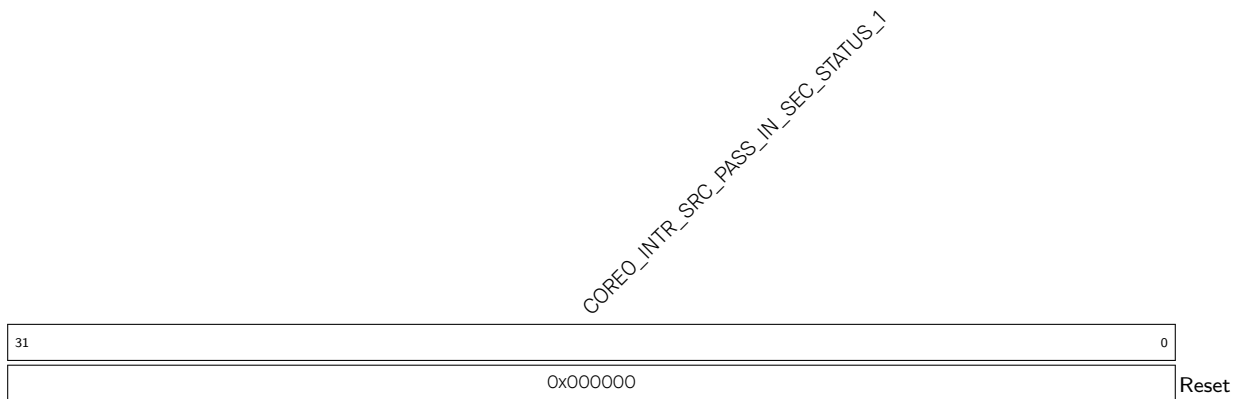


COREO_INTR_SRC_PASS_IN_SEC_STATUS_0 Represents whether interrupt source 0 ~ 31 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

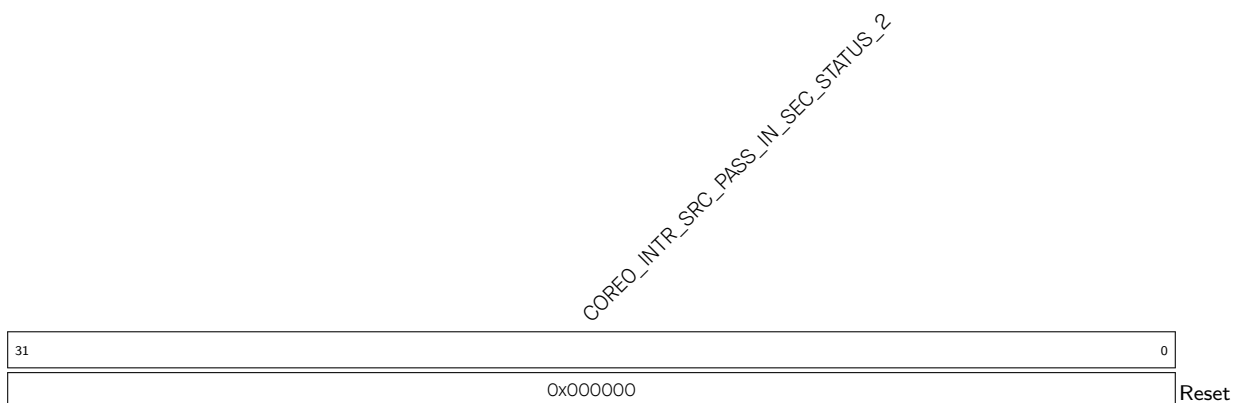
Register 12.139. COREO_INTR_SRC_PASS_IN_SEC_STATUS_1_REG (0x0234)

COREO_INTR_SRC_PASS_IN_SEC_STATUS_1 Represents whether interrupt sources 32 ~ 63 have enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

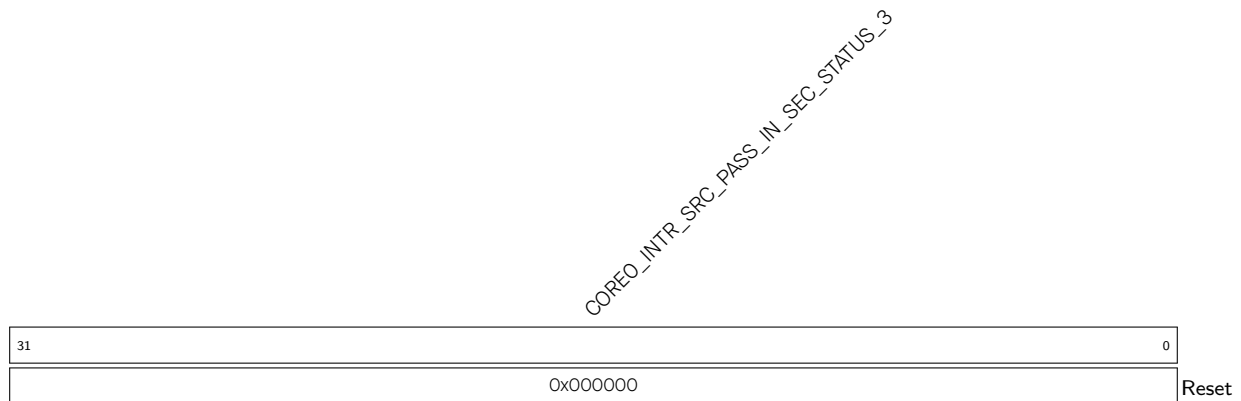
Register 12.140. COREO_INTR_SRC_PASS_IN_SEC_STATUS_2_REG (0x0238)

COREO_INTR_SRC_PASS_IN_SEC_STATUS_2 Represents whether interrupt source 64 ~ 95 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

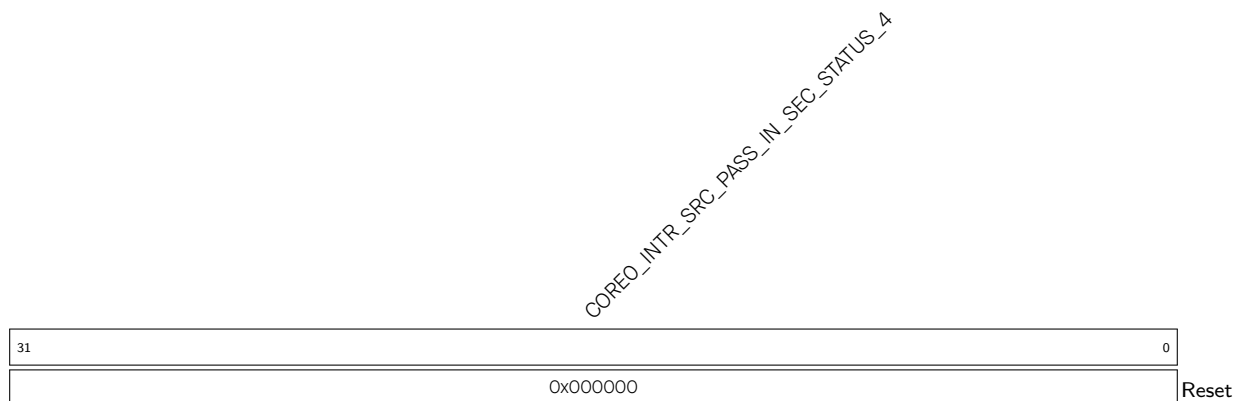
Register 12.141. COREO_INTR_SRC_PASS_IN_SEC_STATUS_3_REG (0x023C)

COREO_INTR_SRC_PASS_IN_SEC_STATUS_3 Represents whether interrupt source 96 ~ 127 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

Register 12.142. COREO_INTR_SRC_PASS_IN_SEC_STATUS_4_REG (0x0240)

COREO_INTR_SRC_PASS_IN_SEC_STATUS_4 Represents whether interrupt source 128 ~ 130 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

Register 12.143. CORE0_INTERRUPT_REG_DATE_REG (0x03FC)

(reserved)				CORE0_INTERRUPT_REG_DATE																									
31	28	27	0																										
0	0	0	0	0x2003020																									Reset

CORE0_INTERRUPT_REG_DATE Version control register. (R/W)

12.6.2 HP CPU1 Interrupt Matrix Registers

Register 12.144. CORE1_LP_RTC_INT_MAP_REG (0x0000)

Register 12.145. CORE1_LP_WDT_INT_MAP_REG (0x0004)

Register 12.146. CORE1_LP_TIMER_REG_O_INT_MAP_REG (0x0008)

Register 12.147. CORE1_LP_TIMER_REG_1_INT_MAP_REG (0x000C)

Register 12.148. CORE1_MB_HP_INT_MAP_REG (0x0010)

Register 12.149. CORE1_MB_LP_INT_MAP_REG (0x0014)

Register 12.150. CORE1_PMU_REG_O_INT_MAP_REG (0x0018)

Register 12.151. CORE1_PMU_REG_1_INT_MAP_REG (0x001C)

Register 12.152. CORE1_LP_ANAPERI_INT_MAP_REG (0x0020)

Register 12.153. CORE1_LP_ADC_INT_MAP_REG (0x0024)

Register 12.154. CORE1_LP_GPIO_INT_MAP_REG (0x0028)

Register 12.155. CORE1_LP_I2C_INT_MAP_REG (0x002C)

Register 12.156. CORE1_LP_I2S_INT_MAP_REG (0x0030)

Register 12.157. CORE1_LP_SPI_INT_MAP_REG (0x0034)

Register 12.158. CORE1_LP_TOUCH_INT_MAP_REG (0x0038)

Register 12.159. CORE1_LP_TSENS_INT_MAP_REG (0x003C)

Register 12.160. CORE1_LP_UART_INT_MAP_REG (0x0040)

Register 12.161. CORE1_LP_EFUSE_INT_MAP_REG (0x0044)

Register 12.162. CORE1_LP_SW_INT_MAP_REG (0x0048)

Register 12.163. CORE1_LP_SYSREG_INT_MAP_REG (0x004C)

Register 12.164. CORE1_LP_HUK_INT_MAP_REG (0x0050)

Register 12.165. CORE1_SYS_ICM_INT_MAP_REG (0x0054)

Register 12.166. CORE1_USB_DEVICE_INT_MAP_REG (0x0058)

Register 12.167. CORE1_ [SDIO_HOST_INT_MAP_REG](#) (0x005C)
Register 12.168. CORE1_ [GDMA_INT_MAP_REG](#) (0x0060)
Register 12.169. CORE1_ [SPI2_INT_MAP_REG](#) (0x0064)
Register 12.170. CORE1_ [SPI3_INT_MAP_REG](#) (0x0068)
Register 12.171. CORE1_ [I2SO_INT_MAP_REG](#) (0x006C)
Register 12.172. CORE1_ [I2S1_INT_MAP_REG](#) (0x0070)
Register 12.173. CORE1_ [I2S2_INT_MAP_REG](#) (0x0074)
Register 12.174. CORE1_ [UHCIO_INT_MAP_REG](#) (0x0078)
Register 12.175. CORE1_ [UART0_INT_MAP_REG](#) (0x007C)
Register 12.176. CORE1_ [UART1_INT_MAP_REG](#) (0x0080)
Register 12.177. CORE1_ [UART2_INT_MAP_REG](#) (0x0084)
Register 12.178. CORE1_ [UART3_INT_MAP_REG](#) (0x0088)
Register 12.179. CORE1_ [UART4_INT_MAP_REG](#) (0x008C)
Register 12.180. CORE1_ [LCD_CAM_INT_MAP_REG](#) (0x0090)
Register 12.181. CORE1_ [ADC_INT_MAP_REG](#) (0x0094)
Register 12.182. CORE1_ [PWMO_INT_MAP_REG](#) (0x0098)
Register 12.183. CORE1_ [PWM1_INT_MAP_REG](#) (0x009C)
Register 12.184. CORE1_ [TWAIO_INT_MAP_REG](#) (0x00A0)
Register 12.185. CORE1_ [TWAI1_INT_MAP_REG](#) (0x00A4)
Register 12.186. CORE1_ [TWAI2_INT_MAP_REG](#) (0x00A8)
Register 12.187. CORE1_ [RMT_INT_MAP_REG](#) (0x00AC)
Register 12.188. CORE1_ [I2CO_INT_MAP_REG](#) (0x00B0)
Register 12.189. CORE1_ [I2C1_INT_MAP_REG](#) (0x00B4)
Register 12.190. CORE1_ [TIMERGRPO_TO_INT_MAP_REG](#) (0x00B8)
Register 12.191. CORE1_ [TIMERGRPO_T1_INT_MAP_REG](#) (0x00BC)
Register 12.192. CORE1_ [TIMERGRPO_WDT_INT_MAP_REG](#) (0x00C0)
Register 12.193. CORE1_ [TIMERGRP1_TO_INT_MAP_REG](#) (0x00C4)
Register 12.194. CORE1_ [TIMERGRP1_T1_INT_MAP_REG](#) (0x00C8)
Register 12.195. CORE1_ [TIMERGRP1_WDT_INT_MAP_REG](#) (0x00CC)
Register 12.196. CORE1_ [LEDC_INT_MAP_REG](#) (0x00D0)
Register 12.197. CORE1_ [SYSTIMER_TARGET0_INT_MAP_REG](#) (0x00D4)
Register 12.198. CORE1_ [SYSTIMER_TARGET1_INT_MAP_REG](#) (0x00D8)
Register 12.199. CORE1_ [SYSTIMER_TARGET2_INT_MAP_REG](#) (0x00DC)
Register 12.200. CORE1_ [AHB_PDMA_IN_CHO_INT_MAP_REG](#) (0x00E0)
Register 12.201. CORE1_ [AHB_PDMA_IN_CH1_INT_MAP_REG](#) (0x00E4)

Register 12.202. CORE1_AHB_PDMA_IN_CH2_INT_MAP_REG (0x00E8)
Register 12.203. CORE1_AHB_PDMA_OUT_CHO_INT_MAP_REG (0x00EC)
Register 12.204. CORE1_AHB_PDMA_OUT_CH1_INT_MAP_REG (0x00F0)
Register 12.205. CORE1_AHB_PDMA_OUT_CH2_INT_MAP_REG (0x00F4)
Register 12.206. CORE1_AXI_PDMA_IN_CHO_INT_MAP_REG (0x00F8)
Register 12.207. CORE1_AXI_PDMA_IN_CH1_INT_MAP_REG (0x00FC)
Register 12.208. CORE1_AXI_PDMA_IN_CH2_INT_MAP_REG (0x0100)
Register 12.209. CORE1_AXI_PDMA_OUT_CHO_INT_MAP_REG (0x0104)
Register 12.210. CORE1_AXI_PDMA_OUT_CH1_INT_MAP_REG (0x0108)
Register 12.211. CORE1_AXI_PDMA_OUT_CH2_INT_MAP_REG (0x010C)
Register 12.212. CORE1_RSA_INT_MAP_REG (0x0110)
Register 12.213. CORE1_AES_INT_MAP_REG (0x0114)
Register 12.214. CORE1_SHA_INT_MAP_REG (0x0118)
Register 12.215. CORE1_ECC_INT_MAP_REG (0x011C)
Register 12.216. CORE1_ECDSA_INT_MAP_REG (0x0120)
Register 12.217. CORE1_KM_INT_MAP_REG (0x0124)
Register 12.218. CORE1_GPIO_INT0_MAP_REG (0x0128)
Register 12.219. CORE1_GPIO_INT1_MAP_REG (0x012C)
Register 12.220. CORE1_GPIO_INT2_MAP_REG (0x0130)
Register 12.221. CORE1_GPIO_INT3_MAP_REG (0x0134)
Register 12.222. CORE1_GPIO_PAD_COMP_INT_MAP_REG (0x0138)
Register 12.223. CORE1_CPU_INT_FROM_CPU_0_MAP_REG (0x013C)
Register 12.224. CORE1_CPU_INT_FROM_CPU_1_MAP_REG (0x0140)
Register 12.225. CORE1_CPU_INT_FROM_CPU_2_MAP_REG (0x0144)
Register 12.226. CORE1_CPU_INT_FROM_CPU_3_MAP_REG (0x0148)
Register 12.227. CORE1_FLASH_MSPI_INT_MAP_REG (0x0150)
Register 12.228. CORE1_CSI_BRIDGE_INT_MAP_REG (0x0154)
Register 12.229. CORE1_DSI_BRIDGE_INT_MAP_REG (0x0158)
Register 12.230. CORE1_CSI_INT_MAP_REG (0x015C)
Register 12.231. CORE1_DSI_INT_MAP_REG (0x0160)
Register 12.232. CORE1_GMII_PHY_INT_MAP_REG (0x0164)
Register 12.233. CORE1_LPI_INT_MAP_REG (0x0168)
Register 12.234. CORE1_PMT_INT_MAP_REG (0x016C)
Register 12.235. CORE1_ETH_MAC_INT_MAP_REG (0x0170)
Register 12.236. CORE1_USB_OTG_INT_MAP_REG (0x0174)

Register 12.237. CORE1_ [USB_OTG_ENDP_MULTI_PROC_INT_MAP_REG](#) (0x0178)

Register 12.238. CORE1_ [JPEG_INT_MAP_REG](#) (0x017C)

Register 12.239. CORE1_ [PPA_INT_MAP_REG](#) (0x0180)

Register 12.240. CORE1_ [CORE0_TRACE_INT_MAP_REG](#) (0x0184)

Register 12.241. CORE1_ [CORE1_TRACE_INT_MAP_REG](#) (0x0188)

Register 12.242. CORE1_ [ISP_INT_MAP_REG](#) (0x0190)

Register 12.243. CORE1_ [I3C_MST_INT_MAP_REG](#) (0x0194)

Register 12.244. CORE1_ [I3C_SLV_INT_MAP_REG](#) (0x0198)

Register 12.245. CORE1_ [USB_OTG11_INT_MAP_REG](#) (0x019C)

Register 12.246. CORE1_ [DMA2D_IN_CHO_INT_MAP_REG](#) (0x01A0)

Register 12.247. CORE1_ [DMA2D_IN_CH1_INT_MAP_REG](#) (0x01A4)

Register 12.248. CORE1_ [DMA2D_OUT_CHO_INT_MAP_REG](#) (0x01A8)

Register 12.249. CORE1_ [DMA2D_OUT_CH1_INT_MAP_REG](#) (0x01AC)

Register 12.250. CORE1_ [DMA2D_OUT_CH2_INT_MAP_REG](#) (0x01B0)

Register 12.251. CORE1_ [PSRAM_MSPI_INT_MAP_REG](#) (0x01B4)

Register 12.252. CORE1_ [HP_SYSREG_INT_MAP_REG](#) (0x01B8)

Register 12.253. CORE1_ [PCNT_INT_MAP_REG](#) (0x01BC)

Register 12.254. CORE1_ [HP_PAU_INT_MAP_REG](#) (0x01C0)

Register 12.255. CORE1_ [HP_PARLIO_RX_INT_MAP_REG](#) (0x01C4)

Register 12.256. CORE1_ [HP_PARLIO_TX_INT_MAP_REG](#) (0x01C8)

Register 12.257. CORE1_ [H264_DMA2D_OUT_CHO_INT_MAP_REG](#) (0x01CC)

Register 12.258. CORE1_ [H264_DMA2D_OUT_CH1_INT_MAP_REG](#) (0x01D0)

Register 12.259. CORE1_ [H264_DMA2D_OUT_CH2_INT_MAP_REG](#) (0x01D4)

Register 12.260. CORE1_ [H264_DMA2D_OUT_CH3_INT_MAP_REG](#) (0x01D8)

Register 12.261. CORE1_ [H264_DMA2D_OUT_CH4_INT_MAP_REG](#) (0x01DC)

Register 12.262. CORE1_ [H264_DMA2D_IN_CHO_INT_MAP_REG](#) (0x01E0)

Register 12.263. CORE1_ [H264_DMA2D_IN_CH1_INT_MAP_REG](#) (0x01E4)

Register 12.264. CORE1_ [H264_DMA2D_IN_CH2_INT_MAP_REG](#) (0x01E8)

Register 12.265. CORE1_ [H264_DMA2D_IN_CH3_INT_MAP_REG](#) (0x01EC)

Register 12.266. CORE1_ [H264_DMA2D_IN_CH4_INT_MAP_REG](#) (0x01F0)

Register 12.267. CORE1_ [H264_DMA2D_IN_CH5_INT_MAP_REG](#) (0x01F4)

Register 12.268. CORE1_ [H264_REG_INT_MAP_REG](#) (0x01F8)

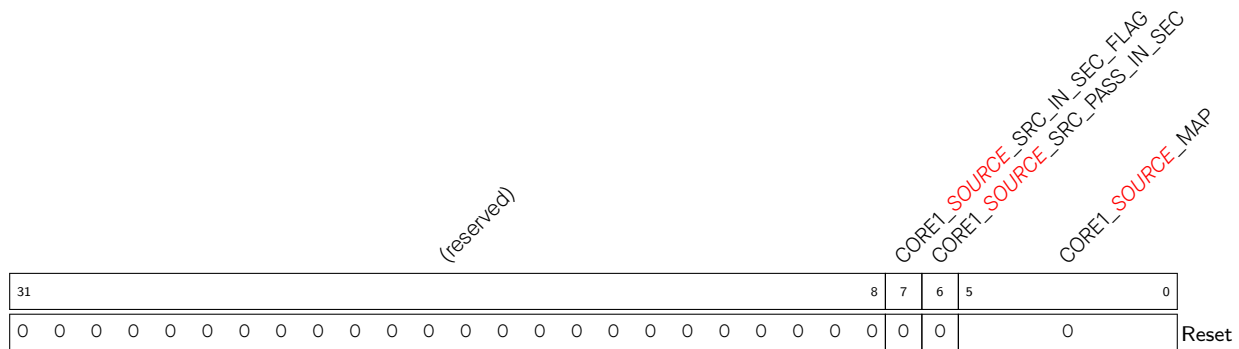
Register 12.269. CORE1_ [ASSIST_DEBUG_INT_MAP_REG](#) (0x01FC)

Register 12.270. CORE1_ [DMA2D_IN_CH2_INT_MAP_REG](#) (0x0214)

Register 12.271. CORE1_ [DMA2D_OUT_CH3_INT_MAP_REG](#) (0x0218)

Register 12.272. CORE1_AXI_PERF_MON_INT_MAP_REG (0x021C)

Register 12.273. CORE1_SOURCE_MAP_REG (0x0000 - 0x01FC, 0x0214 - 0x021C)



CORE1_SOURCE_MAP When **CORE1_SOURCE_SRC_IN_SEC_FLAG** is set to 1, this register can only be written by CPU while in Machine Mode.

Configures this register to map the interrupt signal from the peripheral interrupt sources **Source** to one of CPU1's peripheral interrupts. The valid configuration range is 16 ~ 47.

When set to 0 ~ 15, the corresponding peripheral interrupt source is disabled (recommended value: 0).

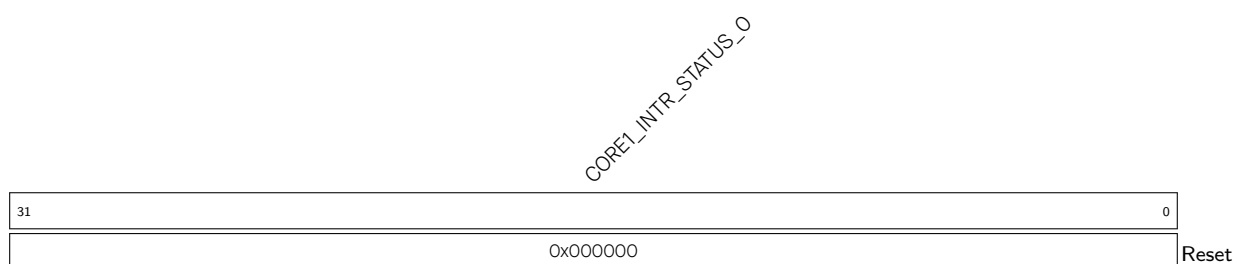
Other values are invalid.

For the list of interrupt sources **Source**, see Table 12.4-1. (R/W)

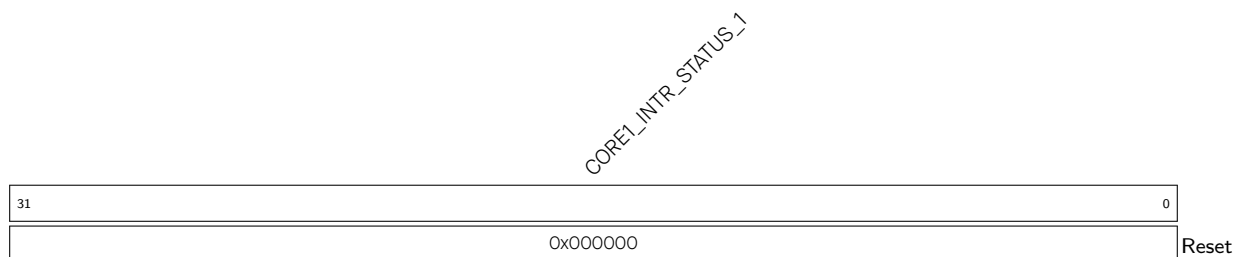
CORE1_SOURCE_SRC_PASS_IN_SEC Write 1 to allow the interrupt signal from interrupt source **Source** to be delegated to CPU1's Machine Mode interrupt. Writable only by the CPU in Machine Mode. (R/W)

CORE1_SOURCE_SRC_IN_SEC_FLAG Write 1 to configure the interrupt signal from interrupt source **Source** as a secure interrupt source (that is, its interrupt mapping can be configured only by the CPU running in Machine Mode). Writable only by the CPU in Machine Mode. (R/W)

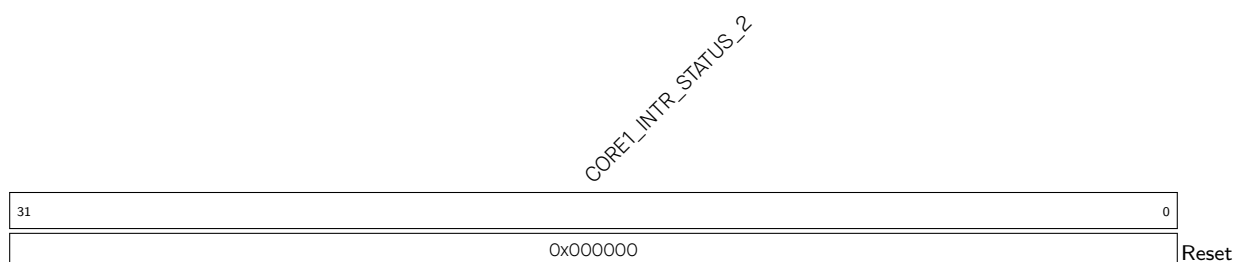
Register 12.274. CORE1_INTR_STATUS_REG_O_REG (0x0200)



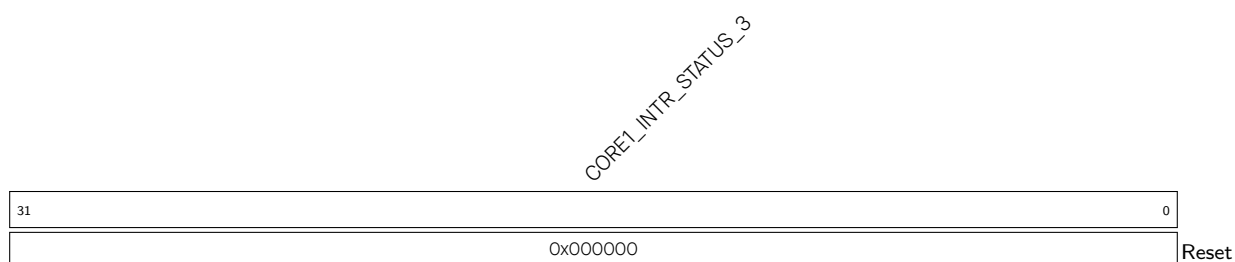
CORE1_INTR_STATUS_O Represents the status of the interrupt sources from 0 ~ 31. Each bit corresponds to one interrupt source. (RO)

Register 12.275. CORE1_INTR_STATUS_REG_1_REG (0x0204)

CORE1_INTR_STATUS_1 Represents the status of the interrupt sources from 32 ~ 63. Each bit corresponds to one interrupt source. (RO)

Register 12.276. CORE1_INTR_STATUS_REG_2_REG (0x0208)

CORE1_INTR_STATUS_2 Represents the status register for interrupt sources from 64 ~ 95. Each bit corresponds to one interrupt source. (RO)

Register 12.277. CORE1_INTR_STATUS_REG_3_REG (0x020C)

CORE1_INTR_STATUS_3 Represents the status of the interrupt sources from 96 ~ 127. Each bit corresponds to one interrupt source. (RO)

Register 12.278. CORE1_CLOCK_GATE_REG (0x0210)

(reserved)																												CORE1_REG_CLK_EN	
31																												1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Reset

CORE1_REG_CLK_EN Configures whether to force interrupt register clock-gate on.

0: No effect

1: Force interrupt register clock-gate on

(R/W)

Register 12.279. CORE1_INTR_STATUS_REG_4_REG (0x0220)

CORE1_INTR_STATUS_4																															
31																															0
0x000000																															
Reset																															

Reset

CORE1_INTR_STATUS_4 Represents the status of the interrupt sources from 128 ~ 130. Each bit corresponds to one interrupt source. (RO)

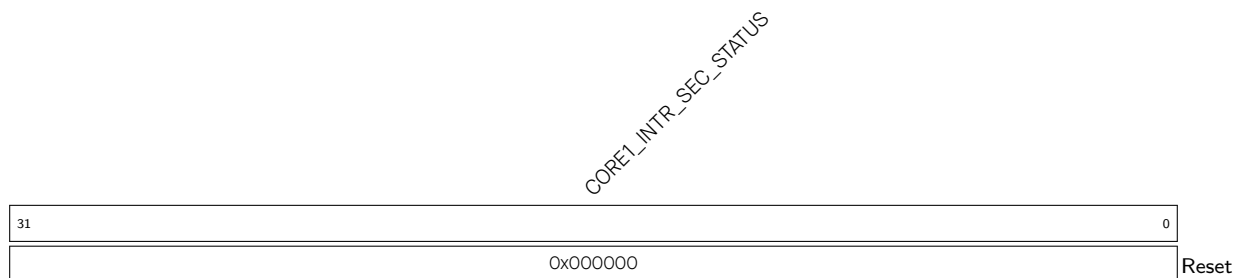
Register 12.280. CORE1_INTR_SIG_IDX_ASSERT_IN_SEC_REG (0x0228)

(reserved)																												CORE1_INTR_SIG_IDX_ASSERT_IN_SEC											
31																												6	5	0									
0 0																												0x000000										Reset	

Reset

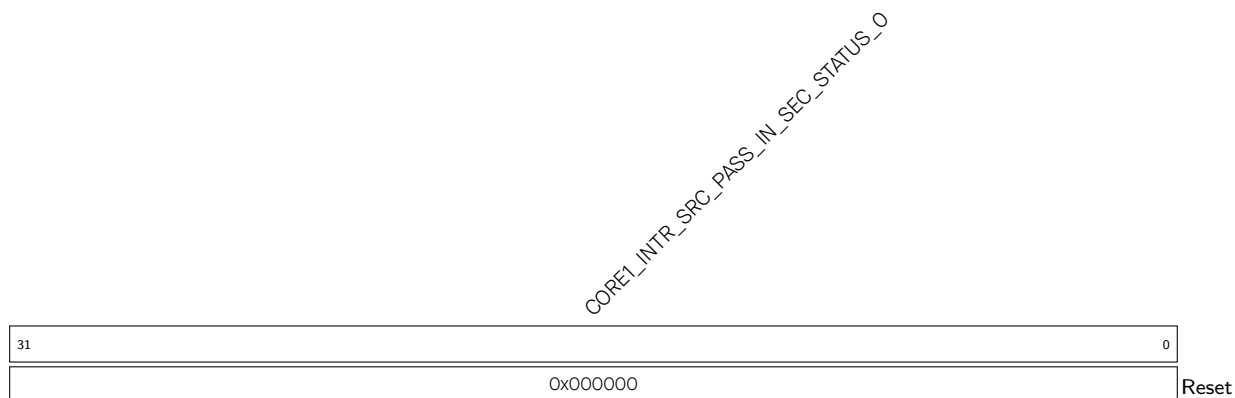
CORE1_INTR_SIG_IDX_ASSERT_IN_SEC Configures which peripheral machine mode interrupt of CPU1 is used as the target for the interrupt delegation function. Valid values range from 16 ~ 47. (R/W)

Register 12.281. CORE1_INTR_SEC_STATUS_REG (0x022C)



CORE1_INTR_SEC_STATUS Represents status of the user mode interrupt signals that have been delegated to the machine mode interrupt when CORE1 is in machine mode. (RO)

Register 12.282. CORE1_INTR_SRC_PASS_IN_SEC_STATUS_0_REG (0x0230)

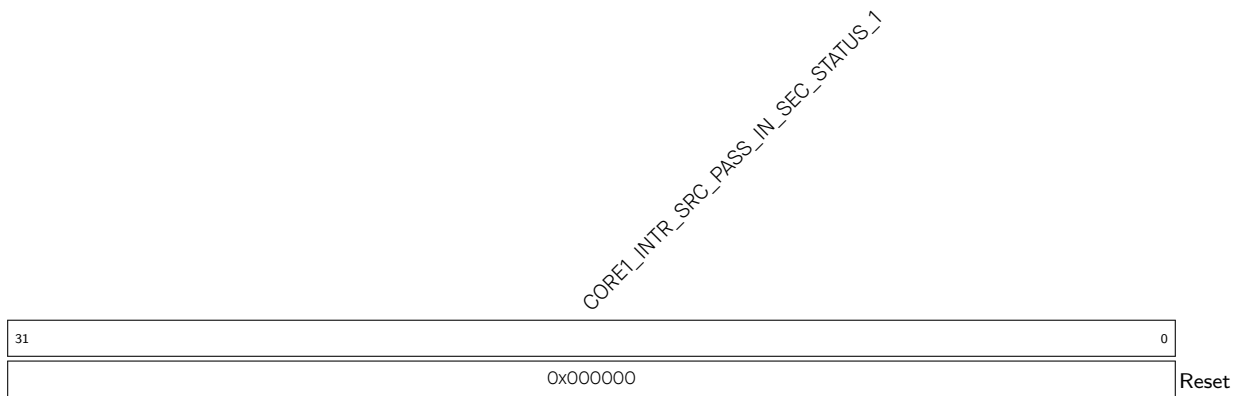


CORE1_INTR_SRC_PASS_IN_SEC_STATUS_0 Represents whether interrupt source 0 ~ 31 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

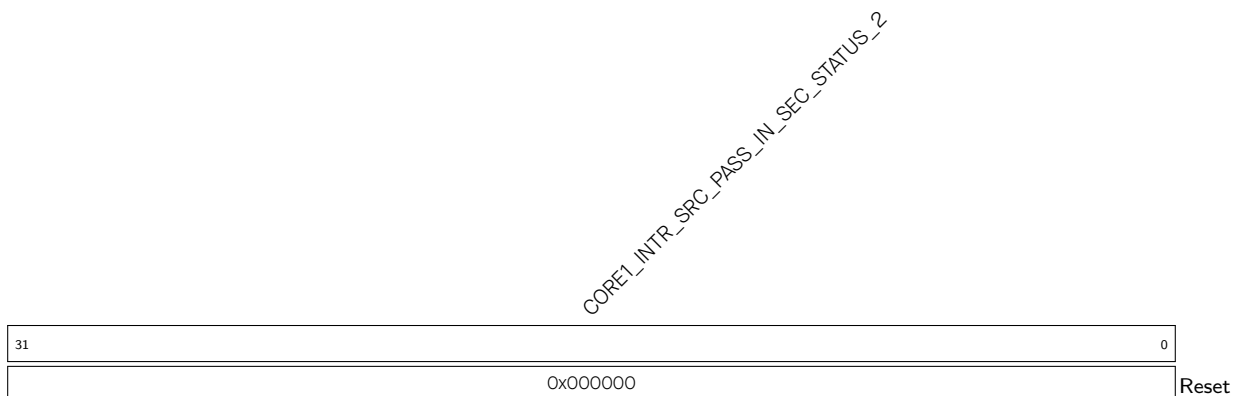
Register 12.283. CORE1_INTR_SRC_PASS_IN_SEC_STATUS_1_REG (0x0234)

CORE1_INTR_SRC_PASS_IN_SEC_STATUS_1 Represents whether interrupt source 32 ~ 63 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

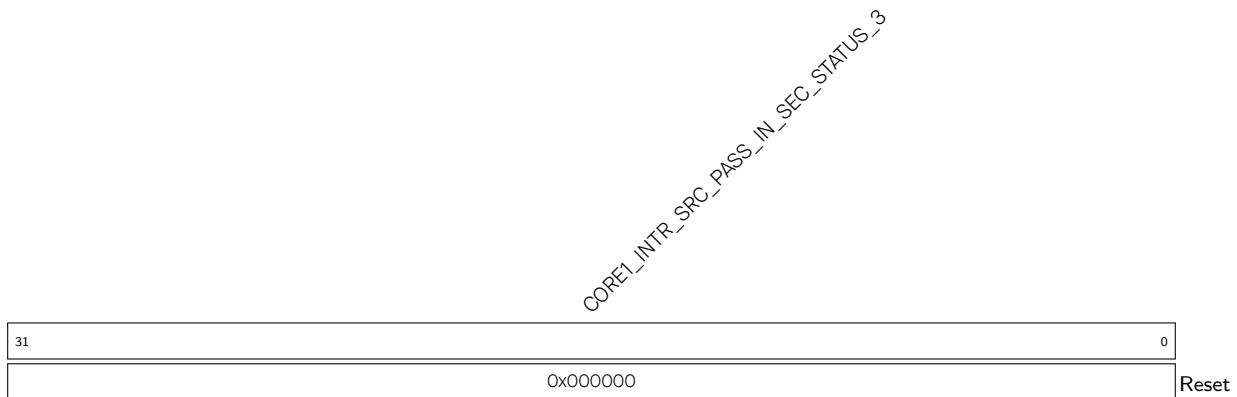
Register 12.284. CORE1_INTR_SRC_PASS_IN_SEC_STATUS_2_REG (0x0238)

CORE1_INTR_SRC_PASS_IN_SEC_STATUS_2 Represents whether interrupt source 64 ~ 95 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

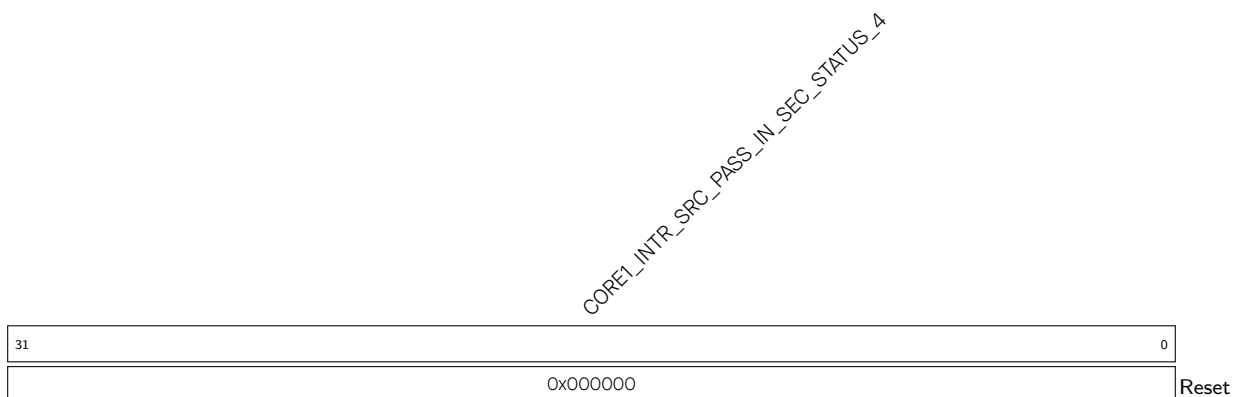
Register 12.285. CORE1_INTR_SRC_PASS_IN_SEC_STATUS_3_REG (0x023C)

CORE1_INTR_SRC_PASS_IN_SEC_STATUS_3 Represents whether interrupt source 96 ~ 127 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

Register 12.286. CORE1_INTR_SRC_PASS_IN_SEC_STATUS_4_REG (0x0240)

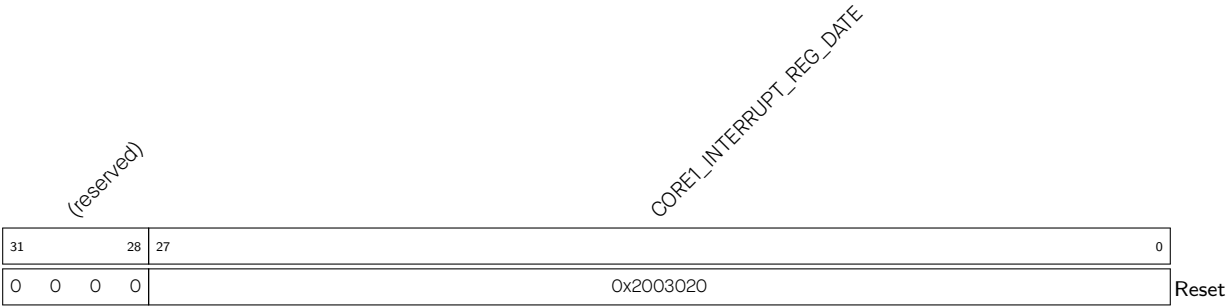
CORE1_INTR_SRC_PASS_IN_SEC_STATUS_4 Represents whether interrupt source 128 ~ 130 has enabled the interrupt delegation function. Each bit corresponds to the enable status of one interrupt source.

0: Corresponding interrupt source has not enabled the interrupt delegation function.

1: Corresponding interrupt source has enabled the interrupt delegation function.

(RO)

Register 12.287. CORE1_INTERRUPT_REG_DATE_REG (0x03FC)



CORE1_INTERRUPT_REG_DATE Version control register. (R/W)